QTM 220: AM

Aronow and Miller plus

Adam Glynn

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November 30, 2022

The expected value [expectation] of an R.V.: the average realization over hypothetical random draws.

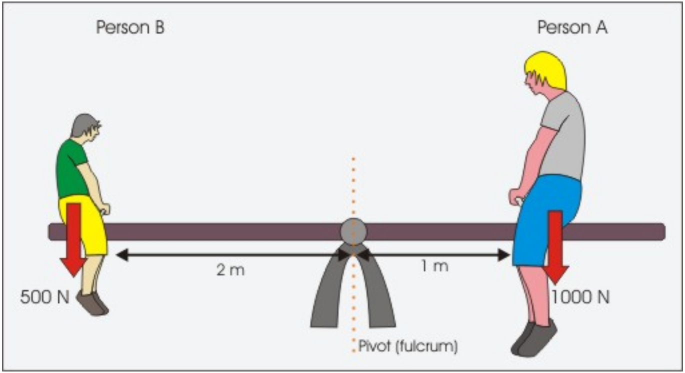
Discrete:

$$\mathrm{E}[X] = \sum_{x \in \mathrm{Supp}(X)} x f(x)$$

Continuous:

$$\mathrm{E}[X] = \int_{-\infty}^{\infty} x f(x) dx$$

What does this mean, physically?


Person B
Person A
2 m
1 m
500 N
1000 N
Pivot (fulcrum)

What do we mean by random draws?

i	1	2	$\dots$	625	
s	Y_1	Y_2	$\dots$	Y_{625}	$\overline{Y}_{625}$
1	1	1	$\dots$	0	.68
2	0	1	$\dots$	1	.70
3	0	0	$\dots$	1	.75
4	0	1	$\dots$	1	.73
5	0	1	$\dots$	1	.69
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
1 mil	0	1	$\dots$	1	.74
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$f_{\cdot}(\cdot)$	Be(.71)	Be(.71)		Be(.71)	$\frac{Bin(625,.71)}{625}$
$\mathrm{E}_{\cdot}(\cdot)$	.71	.71		.71	.71

Beyond binary

i	1	2	...	625	
s	Y_1	Y_2	...	Y_{625}	$\overline{Y}_{625}$
1	35	50	...	22	45
2	28	19	...	65	47
3	33	27	...	45	42
4	⋮	⋮	...	⋮	⋮
5	⋮	⋮	...	⋮	⋮
⋮	⋮	⋮	⋮	⋮	⋮
1 mil	41	53	...	21	46
⋮	⋮	⋮	⋮	⋮	⋮
$f_{\cdot}(\cdot)$	?	?		?	?
$\mathrm{E}_{\cdot}(\cdot)$	45	45		45	?

Important properties:

$$\mathrm{E}[c] = c$$

$$\mathrm{E}[aX] = a\mathrm{E}[X]$$

Prove these assuming X is a discrete RV.

Higher moments about the mean. The *j*th *central* moment:

$$\mu_j = \mathrm{E}\left[(X - \mathrm{E}[X])^j\right].$$

The 2nd *central* moment, the variance:

$$
\begin{aligned}
V(X) &= \mathrm{E}\left[(X - \mathrm{E}[X])^2\right] \\
&= \\
&= \\
&= \\
&= \\
&= \mathrm{E}\left[X^2\right] - \mathrm{E}[X]^2
\end{aligned}
$$

- The average squared deviation from the expected value.
- Expectation of the square minus the square of the expectation.

What does this mean, physically?

The 2nd *central* moment, the variance:

$$\begin{aligned}
V(X) &= \mathrm{E}\left[(X - \mathrm{E}[X])^2\right] \\
&= \mathrm{E}\left[X^2 - 2X\mathrm{E}[X] + \mathrm{E}[X]^2\right] \\
&= \mathrm{E}\left[X^2\right] - 2\mathrm{E}\left[X\mathrm{E}[X]\right] + \mathrm{E}\left[\mathrm{E}[X]^2\right] \\
&= \mathrm{E}\left[X^2\right] - 2\mathrm{E}[X]\mathrm{E}[X] + \mathrm{E}[X]^2 \\
&= \mathrm{E}\left[X^2\right] - 2\mathrm{E}[X]^2 + \mathrm{E}[X]^2 \\
&= \mathrm{E}\left[X^2\right] - \mathrm{E}[X]^2
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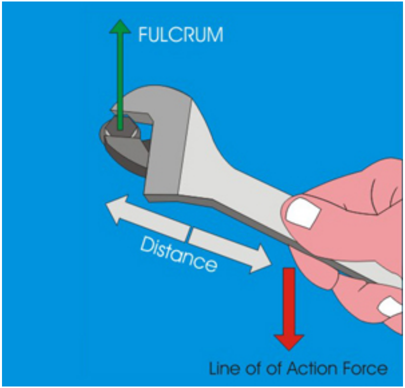
FULCRUM
Distance
Line of of Action Force

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$f_{\cdot}(\cdot)$	Be(.71)	Be(.71)		Be(.71)	$\frac{Bin(625,.71)}{625}$
$\mathrm{E}_{\cdot}(\cdot)$	.71	.71		.71	.71
$V_{\cdot}(\cdot)$	.71(1-.71)	.71(1-.71)		.71(1-.71)	?

Beyond binary

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$f_{\cdot}(\cdot)$	?	?		?	?
$V_{\cdot}(\cdot)$	25	25		25	?

Important properties:

$$V[c] = 0$$

$$V[aX] = a^2 V[X]$$

$$V[c + aX] = a^2 V[X]$$

Prove these assuming X is a discrete RV.

What do these properties mean?

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$V_{\cdot}(\cdot)$	.71(1-.71)	.71(1-.71)		.71(1-.71)	$\frac{625(.71)(1-.71)}{625^2}$

Definition:

Standard deviation: $\sigma(X) = \sqrt{V(X)}$.

Definition: mean squared error (MSE) about a guess c:

$$
\begin{aligned}
\mathrm{E}\left[(X-c)^2\right] &= \\
&= \\
&= V(X) + (\mathrm{E}[X] - c)^2
\end{aligned}
$$

Definition: mean squared error (MSE) about a guess c:

$$\begin{aligned}
\mathrm{E}\left[(X-c)^2\right] &= \mathrm{E}\left[X^2\right] - 2c\mathrm{E}\left[X\right] + c^2 \\
&= \mathrm{E}\left[X^2\right] - \mathrm{E}\left[X\right]^2 + \mathrm{E}\left[X\right]^2 - 2c\mathrm{E}\left[X\right] + c^2 \\
&= V(X) + (\mathrm{E}\left[X\right] - c)^2
\end{aligned}$$

What is the minimum MSE choice of c?

Defining expectation over a vector of RVs.

$$\mathrm{E}\left[(X, Y)\right] = (\mathrm{E}\left[X\right], \mathrm{E}\left[Y\right])$$

Defining an expectation over functions multiple R.V.s.

Let $Z = h(X, Y)$ be a scalar function of the random vector (X, Y) as we proceed.

Discrete:

$$\mathrm{E}\left[Z\right] = \sum_{x \in \mathrm{Supp}\,(X)} \sum_{y \in \mathrm{Supp}\,(Y)} h(x, y) f(x, y)$$

Continuous:

$$\mathrm{E}\left[Z\right] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(x, y) f(x, y) dy dx$$

So this is just a straightforward generalization (working with joint PMFs/PDFs).

Important property: Prove it for discrete.

$$
\begin{aligned}
\mathrm{E}[aX + bY + c] &= \\
&= \\
&= \\
&+ \\
&= a\mathrm{E}[X] + b\mathrm{E}[Y] + c
\end{aligned}
$$

Important property: Prove it for discrete.

$$
\begin{aligned}
\mathrm{E}\left[aX + bY + c\right] &= \sum_{x \in \mathrm{Supp}\,(X)} \sum_{y \in \mathrm{Supp}\,(Y)} h(x, y) f(x, y) \\
&= \sum_{x \in \mathrm{Supp}\,(X)} \sum_{y \in \mathrm{Supp}\,(Y)} (ax + by + c) f(x, y) \\
&= \sum_{x \in \mathrm{Supp}\,(X)} \sum_{y \in \mathrm{Supp}\,(Y)} a \cdot x \cdot f(x, y) \\
&+ \sum_{x \in \mathrm{Supp}\,(X)} \sum_{y \in \mathrm{Supp}\,(Y)} b \cdot y \cdot f(x, y) + c \\
&= a\mathrm{E}\left[X\right] + b\mathrm{E}\left[Y\right] + c
\end{aligned}
$$

What do these properties mean?

i	1	2	...	625	
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1	35	50	...	22	$\frac{1}{2}35 + \frac{1}{2}50$
2	28	19	...	65	$\frac{1}{2}28 + \frac{1}{2}19$
3	33	27	...	45	$\frac{1}{2}33 + \frac{1}{2}27$
4	$\vdots$	$\vdots$	...	$\vdots$	$\vdots$
5	$\vdots$	$\vdots$	...	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
1 mil	41	53	...	21	$\frac{1}{2}41 + \frac{1}{2}53$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$f_{\cdot}(\cdot)$	?	?		?	?
$\mathrm{E}_{\cdot}(\cdot)$	45	45		45	$\frac{1}{2}45 + \frac{1}{2}45$

Generalization of variance: covariance.

$$
\begin{aligned}
C[X, Y] &= \mathrm{E}\left[[X - \mathrm{E}[X]]\,[Y - \mathrm{E}[Y]]\right] \\
&= \\
&= \mathrm{E}[XY] - \mathrm{E}[X]\mathrm{E}[Y].
\end{aligned}
$$

Note:

$$
\begin{aligned}
C[X, X] &= \mathrm{E}[X^2] - \mathrm{E}[X]^2 = V[X] \\
C[a, X] &= 0
\end{aligned}
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&= \mathrm{E}\,[XY] - \mathrm{E}\,[X]\mathrm{E}\,[Y].
\end{aligned}
$$

Note:

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C[X, X] &= \mathrm{E}\,[X^2] - \mathrm{E}\,[X]^2 = V[X] \\
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$$

Covariance rule

$$C[aX + bY, cU + dV] = acC[X, U] + adC[X, V] + bcC[Y, U] + bdC[Y, V]$$

Variance rule in the multivariate case.

$$V[aX + bY] = a^2 V[X] + b^2 V[Y] + 2abC[X, Y].$$

Derive this from the previous rules.

What does this mean for $V[X - Y]$?

Definition: correlation.

$$\rho[X, Y] = \frac{C[X, Y]}{\sqrt{V[X]}\sqrt{V[Y]}}$$

Correlation measures linear dependence and is bounded in $[-1, 1]$. Importantly, for positive b, d:

$$\rho[a + bX, c + dX] = 1$$

$$\rho[a + bX, c - dX] = -1.$$

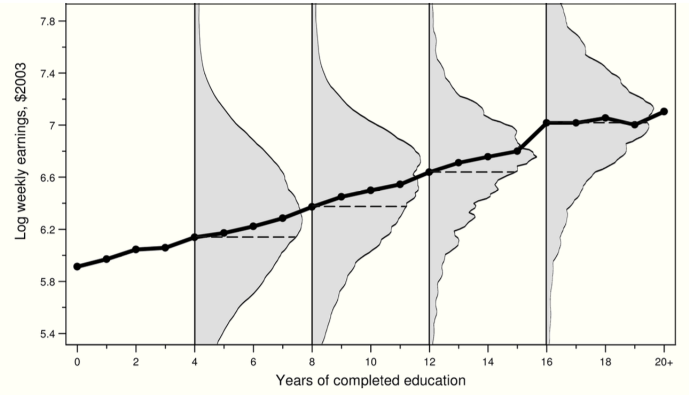
Conditional PMF/PDF transforms a multivariate PMF/PDF

$$f(x, y)$$

to a univariate PMF/PDF

$$f_{Y|X}(y|x) = \frac{f(x, y)}{f(x)}$$

by conditioning on a value of $X = x$.



Log weekly earnings, $2003
7.8
7.4
7
6.6
6.2
5.8
5.4
0
2
4
6
8
10
12
14
16
18
20+
Years of completed education

Discrete:

$$\mathrm{E}[Y|x] = \sum_{y \in \mathrm{Supp}(Y)} y f_{Y|X}(y|x)$$

Continuous:

$$\mathrm{E}[Y|x] = \int_{-\infty}^{\infty} y f_{Y|X}(y|x) dy$$

Everything else carries through, just as though you were operating on a univariate PMF/PDF

E.g.,

$$V[Y|x] = E[Y^2|x] - E[Y|x]^2.$$

Some other rules. Moving from conditional expectations to unconditional expectations.

The *law of iterated expectations*:

$$\mathrm{E}[Y] = \mathrm{E}_X[\mathrm{E}[Y|X]],$$

where E_X refers to the expectation over X.

Draw on board.

Law of iterated expectations directly yields *law of total variance*:

$$V[Y] = \mathrm{E}_X\left[V[Y|X]\right] + V_X\left[\mathrm{E}\left[Y|X\right]\right].$$

Practically speaking, we can decompose the variability of a R.V. Y into the average variability “within" covariate values of X and the average variability “across" covariate values of X.

The conditional expectation function and the best [linear] predictor thereof.

For the random vector (X, Y), the conditional expectation function (CEF) is a function that takes as an input x and returns the conditional expectation of Y given x.

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$$\mathrm{E}[Y|x] = \sum_{y \in \mathrm{Supp}(Y)} y f_{Y|X}(y|x)$$

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$$\mathrm{E}[Y|x] = \int_{-\infty}^{\infty} y f_{Y|X}(y|x) dy$$

The CEF is the best (MMSE) predictor of Y given $X = x$.

I.e., suppose you knew the full joint CDF of Y and X. And then someone gave you a randomly drawn value of X. What would be our guess of Y that would have the lowest MSE?

The answer is given by the CEF. The function $g(X)$ that best approximates Y is the CEF.

But what if we were to restrict ourselves *only* to linear functions? I.e., let $g(X) = a + bX$.

By choosing the a and b that minimize MSE, we are using the best *linear* predictor (BLP) of Y given X.

The BLP, among all functions $g(X) = a + bX$ is given by the constant

$$a = \mathrm{E}[Y] - C[X, Y]/V[X]\mathrm{E}[X]$$

and the slope

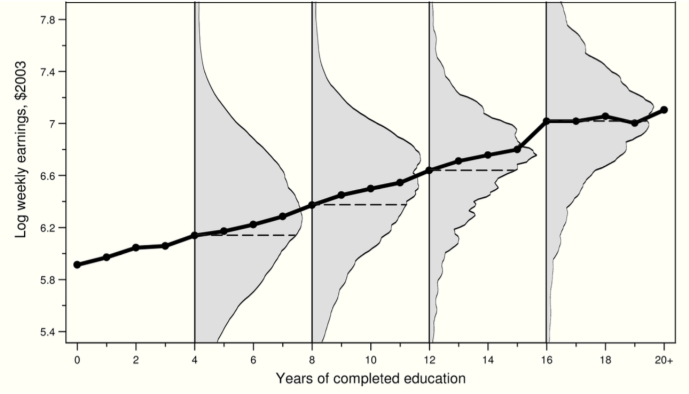
$$b = C[X, Y]/V[X].$$

The proof is very similar to what we did for LS regression solution.

Important corollaries:

- The BLP is also the best linear approximation to the CEF. (I.e., the optimal a and b minimizes $\mathrm{E}\left[\left[\mathrm{E}\left[Y|X\right]-(a+bX)\right]^2\right]$.
- If the CEF *is* linear, the CEF is the BLP.

BLP vs CEF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Sept. 27: What can we learn from a random sample?

Why study random samples?

- Often good design to learn about population averages. E.g., confirmed COVID-19 rates are typically undercounts, and a relatively small randomized sample testing regime may provide better estimates.

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- Often a good provisional model for the first stage of research. E.g., those going to COVID testing centers may approximate a random sample from some subset of the population. Second stage of the research might be to learn about that subset compared to the overall population.
- Transformations used in forecasting can also be conceptualized in terms of random sampling.

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- Equivalent to a random sample with replacement.

Why study i.i.d. samples?

- Equivalent to a random sample with replacement.
- Very good approximation for a random sample without replacement.

Understanding i.i.d.

What does i.i.d. stand for:

- Independent (i.e., observations independent from each other)
- Identically distributed (i.e., all observations drawn from same distribution)

Implications of independence

Y_1 and Y_2 are *independent* if and only if
$f_{Y_1,Y_2}(y_1, y_2) = f_{Y_1}(y_1) \cdot f_{Y_2}(y_2)$, for all (y_1, y_2).

If Y_1 and Y_2 are independent, then:

- Covariance is zero:

$$C(Y_1, Y_2) = 0.$$

- Variances are additive:

$$V(Y_1 + Y_2) = V(Y_1) + V(Y_2).$$

Implications of identically distributed

Y_1 and Y_2 are *independent* if and only if $f_{Y_1,Y_2}(y_1, y_2) = f_{Y_1}(y_1) \cdot f_{Y_2}(y_2)$, for all (y_1, y_2).

Y_1 and Y_2 are *identically distributed* if and only if $f_{Y_1}(y_1) = f_{Y_2}(y_2)$ for all (y_1, y_2).

So, we have n random variables that generate n values.

We can subscript each of these n random variables, as well as aggregate them.

The random vector $\mathbf{Y} = (Y_1, ..., Y_n)'$.

Under the assumption of *random sampling* from F_Y, the collection of values that you see are each produced by an identical random process.

The actual values that we observe are denoted with $\mathbf{y} = (y_1, ..., y_n)'$.

For now, we focus on learning about the population mean ($\mathrm{E}[Y] = \mu$), although to do so, we will need to also learn about the nuisance parameter ($V[Y]$).

We want an estimator with small mean squared error (recall that this is equivalent to squared bias plus variance) and an estimator that allows the production of tight confidence intervals.

A sample statistic *summarizes* those n values. When used to estimate a population parameter, also considered an estimator.

Sample mean:

$$\widehat{\mu}_{sample\ mean} = \overline{Y} = \frac{\sum_{i=1}^{n} Y_i}{n} = \frac{1}{n}\left(Y_1 + ... + Y_n\right).$$

Class of linear combinations:

$$\widehat{\mu}_{\mathbf{c}} = \sum_{i=1}^{n} c_i Y_i = \left(c_1 Y_1 + ... + c_n Y_n\right).$$

Which column is the sampling distribution of the sample mean

i	1	2	$\ldots$	625	
s	Y_1	Y_2	$\ldots$	Y_{625}	$\overline{Y}_{625}$
1	35	50	$\ldots$	22	45
2	28	19	$\ldots$	65	47
3	33	27	$\ldots$	45	42
4	$\vdots$	$\vdots$	$\ldots$	$\vdots$	$\vdots$
5	$\vdots$	$\vdots$	$\ldots$	$\vdots$	$\vdots$
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1 mil	41	53	$\ldots$	21	46
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$f_{\cdot}(\cdot)$	?	?		?	?
$\mathrm{E}_{\cdot}(\cdot)$	45	45		45	?
$V_{\cdot}(\cdot)$	25	25		25	?

Property #1: *The sample mean is unbiased for the population mean.* The expected value of the sample mean, $\mathrm{E}[\overline{Y}] = \mathrm{E}[Y]$.

Proof:

$$
\begin{aligned}
\mathrm{E}[\overline{Y}] &= \mathrm{E}\left[\frac{1}{n}(Y_1 + ... + Y_n)\right] \\
&= \\
&= \\
&= \\
&= \\
&= \mathrm{E}[Y_1].
\end{aligned}
$$

The sample mean is unbiased

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$f_{\cdot}(\cdot)$	?	?		?	?
$\mathrm{E}_{\cdot}(\cdot)$	45	45		45	45
$V_{\cdot}(\cdot)$	25	25		25	?

Within the class of linear combinations, what must be true for unbiasedness?

$$\widehat{\mu}_{\mathbf{c}} = \sum_{i=1}^{n} c_i Y_i = (c_1 Y_1 + ... + c_n Y_n) \, .$$

Property #2: The *sampling variance* of the sample mean,

$$V[\overline{Y}] = \frac{V[Y]}{n}.$$

Proof:

$$\begin{aligned} V[\overline{Y}] &= V\left[\left(\frac{1}{n}Y_1 + ... + \frac{1}{n}Y_n\right)\right] \\ &= \\ &= \\ &= \\ &= \\ &= \frac{1}{n}V[Y]. \end{aligned}$$

Sampling distribution of the sample mean

i	1	2	$\dots$	625	
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$f_{\cdot}(\cdot)$	?	?		?	?
$\mathrm{E}_{\cdot}(\cdot)$	45	45		45	45
$V_{\cdot}(\cdot)$	25	25		25	$\frac{25}{625}$

Within the class of linear combinations, what is the variance in terms of **c**?

$$\widehat{\mu}_{\mathbf{c}} = \sum_{i=1}^{n} c_i Y_i = (c_1 Y_1 + ... + c_n Y_n) \,.$$

What does this expression tell us about minimizing variance?

Learning about population variance

- We've shown that the sample mean is unbiased
- We've shown that the variance of the sample mean is $V[Y]/n$
- We still need to learn something about $V[Y]$

Recall that the “population" variance of a random variable Y:

$$V(Y) = \mathrm{E}\left[Y^2\right] - \mathrm{E}[Y]^2$$

Can we propose an estimator of $V(Y)$?

Put another way:

- Can we propose an estimator of $\mathrm{E}\left[Y^2\right]$?
- Can we propose an estimator of $\mathrm{E}[Y]^2$?

- Can we propose an estimator of $\mathrm{E}\left[Y^2\right]$?
  - Plug-in estimator:

$$\overline{Y^2} = \frac{\sum_{i=1}^{n} Y_i^2}{n}.$$

  - $\mathrm{E}[\overline{Y^2}] = \mathrm{E}\left[Y^2\right]$ (just let $Y^2 = Z$ and it is just $\mathrm{E}[\bar{Z}] = \mathrm{E}[Z]$).

- Can we propose an estimator of $\mathrm{E}[Y]^2$?
  - Plug-in estimator:

$$\overline{Y}^2 = \left[\frac{\sum_{i=1}^{n} Y_i}{n}\right]^2.$$

  - What is it's expectation ($\mathrm{E}[\overline{Y}^2]$)?
  - Recall that $V[\overline{Y}] = \mathrm{E}[\overline{Y}^2] - \mathrm{E}\left[\overline{Y}\right]^2$
  - Simplifies to $\frac{V[Y]}{n} = \mathrm{E}[\overline{Y}^2] - \mathrm{E}\,[Y]^2$
  - Rearranging shows that $\mathrm{E}[\overline{Y}^2] = \mathrm{E}\,[Y]^2 + \frac{V[Y]}{n}$

Therefore, the plug-in estimator of variance,

$$\overline{Y^2} - \overline{Y}^2.$$

is biased:

$$\begin{aligned}
\mathrm{E}\,[\overline{Y^2} - \overline{Y}^2] &= \mathrm{E}\,\left[Y^2\right] - (\mathrm{E}\,[Y]^2 + \frac{V[Y]}{n}) \\
&= (\mathrm{E}\,\left[Y^2\right] - \mathrm{E}\,[Y]^) - \frac{V[Y]}{n}) \\
&= V(Y) - \frac{V[Y]}{n} \\
&= (1 - \frac{1}{n})V(Y) \\
&= \frac{n-1}{n}V(Y)
\end{aligned}$$

However, the bias can be corrected by multiplying the estimator by $\frac{n}{n-1}$:

$$\hat{V}(Y) = \frac{n}{n-1}\left[\overline{Y^2} - \overline{Y}^2\right].$$

with

- $\mathrm{E}\,[\hat{V}(Y)] = \frac{n}{n-1}\mathrm{E}\left[\overline{Y^2} - \overline{Y}^2\right] = \frac{n}{n-1}\frac{n-1}{n}V(Y)$

This is what R computes when you type `var(.)`.

Sampling distribution of the sample variance (fill in for biased sample variance)

i	1	2	$\ldots$	625			
s	Y_1	Y_2	$\ldots$	Y_{625}	$\overline{Y}_{625}$	$\hat{V}(Y)$	$\overline{Y^2} - \overline{Y}^2$
1	35	50	$\ldots$	22	45	20	
2	28	19	$\ldots$	65	47	17	
3	33	27	$\ldots$	45	42	38	
4	$\vdots$	$\vdots$	$\ldots$	$\vdots$	$\vdots$		
5	$\vdots$	$\vdots$	$\ldots$	$\vdots$	$\vdots$		
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$		
1 mil	41	53	$\ldots$	21	46	32	
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$		
$f_{\cdot}(\cdot)$	?	?		?	?		
$\mathrm{E}_{\cdot}(\cdot)$	45	45		45	45	25	
$V_{\cdot}(\cdot)$	25	25		25	$\frac{25}{625}$		

Summary of “Small” Sample Properties

- MSE $= E[(\hat{\mu} - \mu)^2]$
- Bias $= E[(\hat{\mu} - \mu)]$
- Variance $= E[(\hat{\mu} - E[\hat{\mu}])^2]$

$$\underbrace{E[(\hat{\mu} - \mu)^2]}_{MSE} = \underbrace{E[(\hat{\mu} - \mu)]^2}_{Bias\ squared} + \underbrace{E[(\hat{\mu} - E[\hat{\mu}])^2]}_{Variance}$$

Oct. 4: Asymptotic (i.e., large sample) results and confidence intervals

Asymptotic properties of estimators

- Consistency (estimator converges in probability to the population parameter).
- *MSE* $\to 0$ (stronger than consistency, but more intuitive)
- Asymptotic normality (sampling distribution converges in distribution to normal distribution)

Property # 3: Given finite $\mathrm{E}[Y]$ and finite $V(Y) > 0$, then for any $k > 0$,

$$\Pr\left(|\overline{Y} - \mathrm{E}[Y]| \geq k\right) \leq \frac{V[Y]}{k^2 n}.$$

Proof follows directly from Chebyshev's inequality.

Property # 4. [Stated loosely.] *The sample mean is consistent for the population mean.* Given finite $\mathrm{E}[Y]$ and finite $V(Y) > 0$, then as $n \to \infty$,

$$\Pr\left(|\overline{Y} - \mathrm{E}[Y]| \geq k\right) \to 0.$$

Or, equivalently,

$$\overline{Y} \xrightarrow{p} \mathrm{E}[Y],$$

where $\xrightarrow{p}$ is read as “convergence in probability." This is a form of the *weak law of large numbers* (WLLN).

Consistency of $\overline{Y}$ for $\mathrm{E}[Y]$

The WLLN states that

$$\overline{Y} \xrightarrow{p} \mathrm{E}[Y],$$

If we think of $\mathrm{E}[Y]$ as μ and $\overline{Y}$ as $\hat{\mu}$, then this is equivalent to saying that $\hat{\mu}$ is consistent for μ

Easier version based on MSE

$$MSE \to 0 \implies \hat{\theta} \to_p \theta$$

Implies that it is sufficient to show that Bias and Variance go to zero.

Consistency

With the person next to you, explain how you know that

$$\overline{Y} \rightarrow_p \mu$$

Another useful property

(A consequence of the) Continuous Mapping Theorem, stated loosely. Continuous functions are limit-preserving.

For continuous $f(.)$, if $Y \xrightarrow{p} c$, then $f(Y) \xrightarrow{p} f(c)$.

Additionally, for continuous $f(b, c)$ if $X \xrightarrow{p} b$ and $Y \xrightarrow{p} c$, then $f(X, Y) \xrightarrow{p} f(b, c)$

Consistency

With the person next to you, are the following consistent?

- $\overline{Y^2} - \overline{Y}^2$ for σ^2
- $\hat{V}[Y] = \frac{n}{n-1}(\overline{Y^2} - \overline{Y}^2)$ for σ^2

Explain why?

Inconsistent Examples

With the person next to you, come up with at least one example of an estimator for μ that has the following properties:

- unbiased for μ but inconsistent
- variance goes to zero, but inconsistent for μ

Confidence Intervals

Important definition. The *standard error* of the sample mean:

$$\sqrt{V(\overline{Y})}.$$

(This is literally the standard deviation of the sampling distribution of the sample mean.)

An *estimate* of the standard error of the sample mean:

$$\sqrt{\hat{V}(\overline{Y})}.$$

Not unbiased (due to Jensen's inequality), though it is "consistent."

When you see "standard error" in a paper, this is what they mean (an estimate thereof). In theory, it is possible to estimate the sampling variance of the standard error of the sample mean, though.

What is the standard error of the sample mean and what is the estimated standard error of the sample mean?

i	1	2	...	625		
s	Y_1	Y_2	...	Y_{625}	$\overline{Y}_{625}$	$\hat{V}(Y)$
1	35	50	...	22	45	20
2	28	19	...	65	47	17
3	33	27	...	45	42	38
4	⋮	⋮	...	⋮	⋮	
5	⋮	⋮	...	⋮	⋮	
⋮	⋮	⋮	⋮	⋮	⋮	
1 mil	41	53	...	21	46	32
⋮	⋮	⋮	⋮	⋮	⋮	
$f.(\cdot)$	?	?		?	?	
$\mathrm{E}.(\cdot)$	45	45		45	45	25
$V.(\cdot)$	25	25		25	$\frac{25}{625}$	

It is therefore possible to estimate the average of a random variable (the population mean) and estimate our uncertainty (using an estimated SE).

However, we may want to know how likely it would be (across different random samples) for an interval of $\overline{Y} \pm 2$ SEs to contain $E[Y]$.

The central limit theorem provides an approach.

The central limit theorem (CLT) for the sample mean.

In words: if n is large, the sampling distribution of the sample mean ($\overline{Y}$) is approximately normal, regardless of the distribution of Y.

In sampling n times from any R.V. Y with finite $\mathrm{E}[Y] = \mu$ and finite $V(Y) = \sigma^2$, then the standardized sample mean

$$Z = \sqrt{n}(\overline{Y} - \mu)/\sigma$$

converges in distribution as $n \to \infty$ to

$$\mathcal{N}(0, 1),$$

where $\mathcal{N}$ denotes the normal distribution.

I am not going to offer a proof here. The simplest “almost-proof" is given by the de Moivre–Laplace theorem (see the Wikipedia page).

A confidence interval for $E[Y]$, loosely speaking, is an *interval estimate* that covers the true $E[Y]$ with a given probability.

I.e., Suppose we were to repeatedly draw from $F_{\overline{Y}}$ and compute a (valid) 95% confidence interval for $E[Y]$. 95% percent of the time, this interval will encompass $E[Y]$.

Suppose we have:

- An asymptotically normal estimator of $E[Y]$: $\overline{Y}$
- A consistent estimator of $nV(\overline{Y})$: $n\hat{V}(\overline{Y})$

An approximate 95% confidence interval for $\mathrm{E}[Y]$ would be:

$$\left(\overline{Y} - 1.96\sqrt{\hat{V}(\overline{Y})}\,,\, \overline{Y} + 1.96\sqrt{\hat{V}(\overline{Y})}\right).$$

An approximate 95% confidence interval for $\mathrm{E}[Y]$ would be:

$$\left(\overline{Y} - 1.96\sqrt{\frac{\hat{V}(Y)}{n}},\ \overline{Y} + 1.96\sqrt{\frac{\hat{V}(Y)}{n}}\right).$$

i	1	2	...	625		MoE	95% CI
s	Y_1	Y_2	...	Y_{625}	$\overline{Y}_{625}$	$1.96 \cdot \sqrt{\frac{\hat{V}(Y)}{625}}$	$\overline{Y}_{625} \pm MoE$
1	35	50	...	22	45		
2	28	19	...	65	47		
3	33	27	...	45	42		
4	$\vdots$	$\vdots$	...	$\vdots$	$\vdots$		
5	$\vdots$	$\vdots$	...	$\vdots$	$\vdots$		
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$		
1 mil	41	53	...	21	46		
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$		
$f_{.}(\cdot)$	?	?		?	?		
$\mathrm{E}_{.}(\cdot)$	45	45		45	45		
$V_{.}(\cdot)$	25	25		25	$\frac{25}{625}$		

Table: *Sampling Distributions for $\overline{Y}_{625}$, the Margin-of-Error, and the 95% Confidence Interval*

i	1	2	$\dots$	625		MoE	95% CI
s	Y_1	Y_2	$\dots$	Y_{625}	$\overline{Y}_{625}$	$1.96 \cdot \sqrt{\frac{\hat{V}(Y)}{625}}$	$\overline{Y}_{625} \pm MoE$
1	35	50	$\dots$	22	45	.198	
2	28	19	$\dots$	65	47	.195	
3	33	27	$\dots$	45	42	.204	
4	$\vdots$	$\vdots$	$\dots$	$\vdots$	$\vdots$	.201	
5	$\vdots$	$\vdots$	$\dots$	$\vdots$	$\vdots$	.205	
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	
1 mil	41	53	$\dots$	21	46	.197	
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	
$f_{\cdot}(\cdot)$	?	?		?	?		
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i	1	2	...	625		MoE	95% CI
s	Y_1	Y_2	...	Y_{625}	$\overline{Y}_{625}$	$1.96 \cdot \sqrt{\frac{\hat{V}(Y)}{625}}$	$\overline{Y}_{625} \pm MoE$
1	35	50	...	22	45	.198	[44.83, 45.60]
2	28	19	...	65	47	.195	[44.46, 45.22]
3	33	27	...	45	42	.204	[44.61, 45.41]
4	⋮	⋮	...	⋮	⋮	.201	[45.11, 45.88]
5	⋮	⋮	...	⋮	⋮	.205	[44.54, 45.31]
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
1 mil	41	53	...	21	46	.197	[44.64, 45.46]
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
$f.(\cdot)$	?	?		?	?		
$\mathrm{E}.(\cdot)$	45	45		45	45		
$V.(\cdot)$	25	25		25	$\frac{25}{625}$		

Table: *Sampling Distributions for $\overline{Y}_{625}$, the Margin-of-Error, and the 95% Confidence Interval*

A quick note on hypothesis testing

Recall that the 95% CI we estimated was $[44.83, 45.60]$. This allows us to reject at the $\alpha = 5\%$ level all null hypotheses of the following form:

$$H_0 : \mu = c$$
$$H_1 : \mu \neq c$$

for $c < 43.83$ or $c > 45.60$. For example, we can reject the null hypothesis that this neighborhood has an average income equal to the U.S. median of 70k.

A loose proof for the asymptotic validity of confidence intervals under a normal approximation.

Suppose n is large.

Define the R.V.

$$Z = \frac{\overline{Y} - E[Y]}{\sqrt{V(\overline{Y})}},$$

where Z is distributed $\mathcal{N}(0, 1)$ (by asymptotic normality).

Plugging in the *estimated* standard error.

Define the R.V.

$$Z' = \frac{\overline{Y} - E[Y]}{\sqrt{V(\overline{Y})}} \frac{\sqrt{nV(\overline{Y})}}{\sqrt{n\hat{V}(\overline{Y})}} = \frac{\overline{Y} - E[Y]}{\sqrt{\hat{V}(\overline{Y})}}$$

where Z' is distributed $\mathcal{N}(0, 1)$ (by asymptotic normality, the consistency of variance estimator, and Slutsky's Theorem).

Bootstrapping

1. Take a sample of size n with replacement from your sample
2. Calculate $\mu^{(b=1)}$ from that bootstrap sample
3. Repeat for $b = 1, ...B$ where B is big
4. Do one of the following:
    - Calculate bootstrapped variance $\frac{1}{B-1}\sum_{b=1}^{B}(\mu^{(b)} - \overline{\mu^{(b)}})^2$
    - Put $\{\mu^{(b)} : b = 1, ..., B\}$ in order and report the $\alpha/2$ and $1 - \alpha/2$ percentiles
    - ...

Go to bootstrapping code.

Why does bootstrapping work? (informal part 1)

- Empirical CDF ($\hat{F}(y) = \overline{I(Y \leq y)}$) is a collection of sample means/proportions for all y.

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Why does bootstrapping work? (informal part 1)

- Empirical CDF ($\hat{F}(y) = \overline{I(Y \leq y)}$) is a collection of sample means/proportions for all y.
- Therefore, the empirical CDF converges to population CDF ($F(y)$) for all y (AM pp. 116-117)
- If the empirical CDF is close enough to the population CDF, then a resample with replacement from the sample is like a sample from the population

Why does bootstrapping work? (informal part 2)

- Population parameters (e.g., the population mean) can be written as functionals of the CDF ($E[Y] = \int ydF(y)$)

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- Plug in estimators (e.g., the sample mean) can be thought of as functionals of the empirical CDF ($\overline{Y} = \int y d\hat{F}(y)$)
- If the functional is well behaved (e.g., the mean) and not sensitive to small changes in the CDF (e.g., the max is poorly behaved), then the plug-in estimator will converge to the population parameter in many ways (AM pg. 120)

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- Population parameters (e.g., the population mean) can be written as functionals of the CDF ($E[Y] = \int ydF(y)$)
- Plug in estimators (e.g., the sample mean) can be thought of as functionals of the empirical CDF ($\overline{Y} = \int yd\hat{F}(y)$)
- If the functional is well behaved (e.g., the mean) and not sensitive to small changes in the CDF (e.g., the max is poorly behaved), then the plug-in estimator will converge to the population parameter in many ways (AM pg. 120)
- And the plug-in estimators from resamples with replacement will converge toward a normal distribution in the same manner (with enough sample size).

Oct. 18: Linear regression can be seen a way of estimating the BLP using the data in your sample.

What have we been doing?

We have been estimating population parameters (e.g., $E[Y] = \mu$ and $V[Y] = \sigma^2$) for a single variable (i.e., univariate) from an i.i.d. sample.

Those estimators are a function of the sample

- $\hat{\mu} = \overline{Y}$ for μ
- $\hat{\sigma}^2_{biased} = \overline{Y^2} - \overline{Y}^2$ or $\hat{\sigma}^2 = \frac{n}{n-1}(\overline{Y^2} - \overline{Y}^2)$ for σ^2
- $[\hat{\mu} - 1.96\frac{\hat{\sigma}^2}{n}, \hat{\mu} + 1.96\frac{\hat{\sigma}^2}{n}]$

We have considered properties of these estimators:

- Bias
- Variance
- Consistency
- Asymptotic Normality
- Asymptotic Coverage

Epstein and Mershon SCOTUS data

- Data on 27 justices from the Warren, Burger, and Rehnquist courts (can be interpreted as a census)

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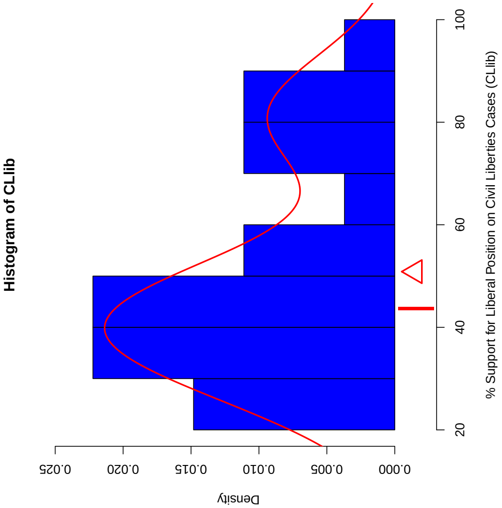
Epstein and Mershon SCOTUS data

- Data on 27 justices from the Warren, Burger, and Rehnquist courts (can be interpreted as a census)
- Percentage of votes in liberal direction for each justice in a number of issue areas
- Segal-Cover scores for each justice
- Party of appointing president

Mean (μ) judicial support for liberal position



What are we doing now?

We will be estimating conditional population parameters ($E[Y|X = x] \approx \beta_0 + \beta_1 x$) and contrasts of population parameters ($E_x(E[Y|X = x + 1] - E[Y|X = x]) \approx \beta_1$) from an i.i.d. sample.

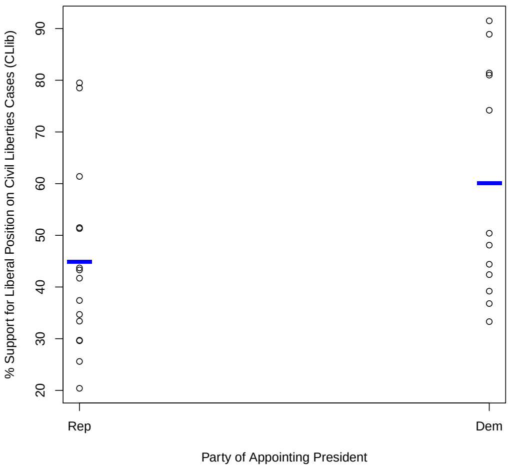
For example, we will estimate β_1 with

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)y_i}{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)^2}$$

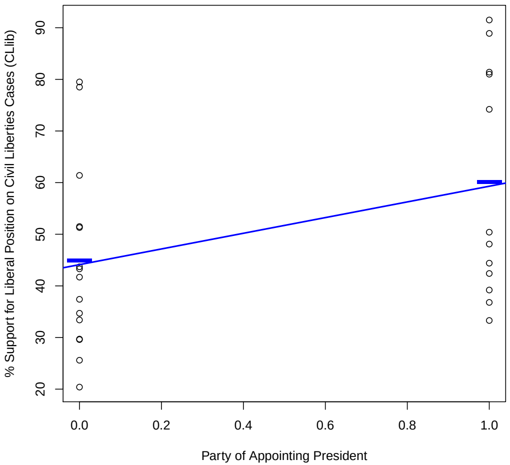
and consider properties of this estimator:

- Bias
- Variance
- Consistency
- Asymptotic Normality

CLlib Mean Conditional on Party

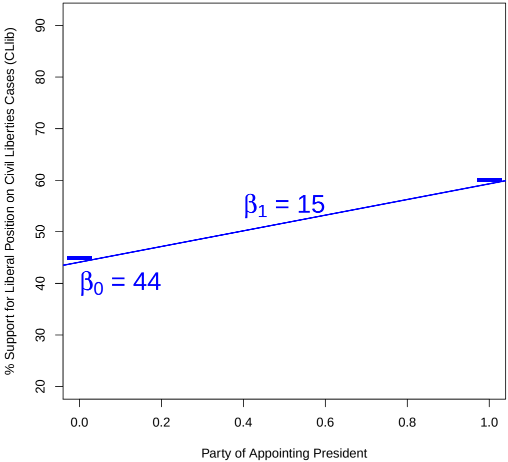


BLP of CLlib



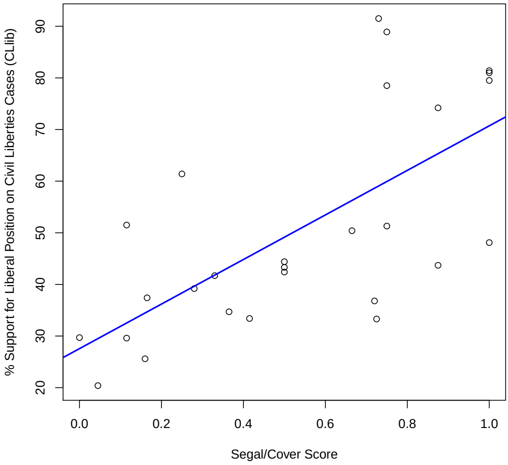
% Support for Liberal Position on Civil Liberties Cases (CLlib)
20
30
40
50
60
70
80
90
0.0
0.2
0.4
0.6
0.8
1.0
Party of Appointing President

BLP of CLlib

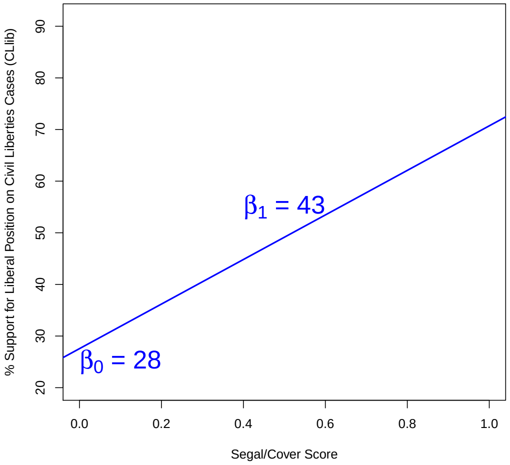


How are the BLP and the CEF related here?

BLP for CLlib Conditional on SCscore



BLP for CLlib Conditional on SCscore



Reminder: where do our estimators come from?

- $\hat{\mu} = \overline{Y}$
- $\hat{\sigma}^2_{biased} = \overline{Y^2} - \overline{Y}^2$
- $\hat{\sigma}^2 = \frac{n}{n-1}(\overline{Y^2} - \overline{Y}^2)$

Reminder: where do our estimators come from?

- $\hat{\mu} = \overline{Y}$
- $\hat{\sigma}^2_{biased} = \overline{Y^2} - \overline{Y}^2$
- $\hat{\sigma}^2 = \frac{n}{n-1}(\overline{Y^2} - \overline{Y}^2)$

what about

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)y_i}{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)^2}$$

Toward a Plug-in Estimator

The BLP of Y given X is $g(X) = \beta_0 + \beta_1 X$, where

$$\beta_0 = \mathrm{E}[Y] - \frac{C[X, Y]}{V[X]}\mathrm{E}[X]$$

and the slope

$$\beta_1 = \frac{C[X, Y]}{V[X]}.$$

Toward a Plug-in Estimator

The BLP of Y given X is $g(X) = \beta_0 + bX$, where

$$\beta_0 = \mathrm{E}\left[Y\right] - \frac{\mathrm{E}\left[XY\right] - \mathrm{E}\left[X\right]\mathrm{E}\left[Y\right]}{\mathrm{E}\left[X^2\right] - \mathrm{E}\left[X\right]^2}\mathrm{E}\left[X\right]$$

and the slope

$$\beta_1 = \frac{\mathrm{E}\left[XY\right] - \mathrm{E}\left[X\right]\mathrm{E}\left[Y\right]}{\mathrm{E}\left[X^2\right] - \mathrm{E}\left[X\right]^2}$$

A plug-in *estimator* of the BLP:

$$\hat{\beta}_0 = \overline{Y} - \frac{\overline{XY} - \overline{X} \cdot \overline{Y}}{\overline{X^2} - \overline{X}^2} \overline{X}$$

and the slope

$$\begin{aligned}
\hat{\beta}_1 &= \frac{\overline{XY} - \overline{X} \cdot \overline{Y}}{\overline{X^2} - \overline{X}^2} \\
&= \frac{\sum_{i=1}^n (x_{i1} - \bar{x}_1) y_i}{\sum_{i=1}^n (x_{i1} - \bar{x}_1)^2}
\end{aligned}$$

Unbiasedness of $\hat{\beta}_1$ for β_1

- If we write $y_i = \beta_0 + \beta_1 x_{1i} + \epsilon_i$. Replace y_i. Then,

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)y_i}{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)^2}$$

$$=$$

$$=$$

- If we write $y_i = \beta_0 + \beta_1 x_{1i} + \epsilon_i$. Replace y_i. Then,

$$
\begin{aligned}
\hat{\beta}_1 &= \frac{\sum_{i=1}^{n}(x_{i1} - \bar{x_1})y_i}{\sum_{i=1}^{n}(x_{i1} - \bar{x_1})^2} \\
&= \frac{\sum_{i=1}^{n}(x_{i1} - \bar{x_1})(\beta_0 + \beta_1 x_{i1} + \epsilon_i)}{\sum_{i=1}^{n}(x_{i1} - \bar{x_1})^2} \\
&=
\end{aligned}
$$

- If we write $y_i = \beta_0 + \beta_1 x_{1i} + \epsilon_i$. Replace y_i. Then,

$$
\begin{aligned}
\hat{\beta}_1 &= \frac{\sum_{i=1}^{n}(x_{i1} - \bar{x_1})y_i}{\sum_{i=1}^{n}(x_{i1} - \bar{x_1})^2} \\
&= \frac{\sum_{i=1}^{n}(x_{i1} - \bar{x_1})(\beta_0 + \beta_1 x_{i1} + \epsilon_i)}{\sum_{i=1}^{n}(x_{i1} - \bar{x_1})^2} \\
&= \frac{\sum_{i=1}^{n}(x_{i1} - \bar{x_1})\beta_0}{\sum_{i=1}^{n}(x_{i1} - \bar{x_1})^2} + \frac{\sum_{i=1}^{n}(x_{i1} - \bar{x_1})\beta_1 x_{i1}}{\sum_{i=1}^{n}(x_{i1} - \bar{x_1})^2} + \frac{\sum_{i=1}^{n}(x_{i1} - \bar{x_1})\epsilon_i}{\sum_{i=1}^{n}(x_{i1} - \bar{x_1})^2}
\end{aligned}
$$

- Therefore we have

$$\hat{\beta}_1 = \beta_1 + \frac{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)\epsilon_i}{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)^2}$$

Take expectation conditional on the all the $x_{i1}, i = \ldots, n$,

$$E\left[\hat{\beta}_1 | x_{i1}, x_{i2}, i = 1, \ldots, n\right]$$

$$=$$

$$=$$

- Therefore we have

$$\hat{\beta}_1 = \beta_1 + \frac{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)\epsilon_i}{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)^2}$$

Take expectation conditional on the all the $x_{i1}, i = \ldots, n$,

$$\begin{aligned} & E\left[\hat{\beta}_1 | x_{i1}, x_{i2}, i = 1, \ldots, n\right] \\ &= \beta_1 + \frac{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1) E\left[\epsilon_i | x_{i1}, x_{i2}, i = 1, \ldots, n\right]}{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)^2} \\ &= \end{aligned}$$

- Therefore we have

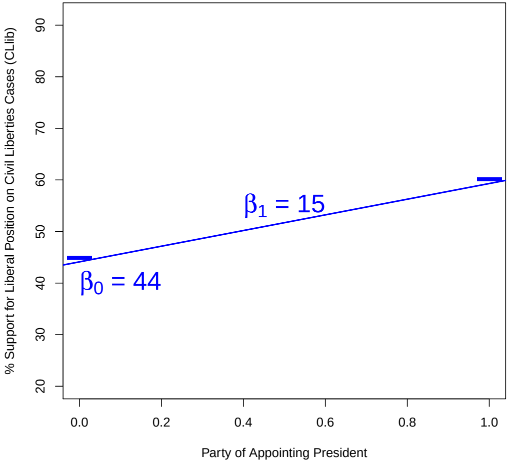
$$\hat{\beta}_1 = \beta_1 + \frac{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)\epsilon_i}{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)^2}$$

Take expectation conditional on the all the $x_{i1}, i = \ldots, n$,

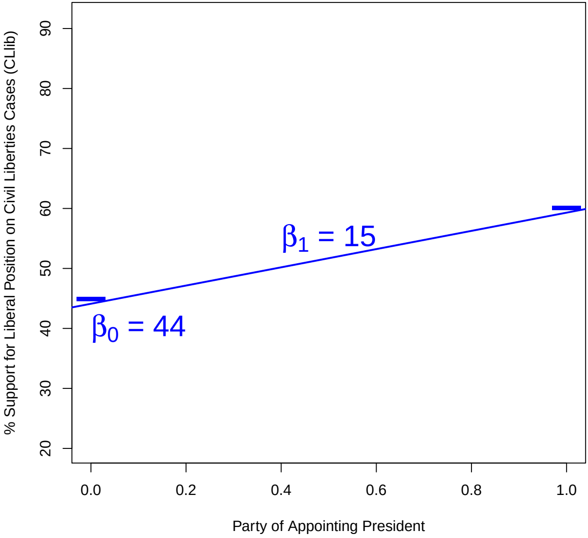
$$\begin{aligned} & E\left[\hat{\beta}_1 | x_{i1}, x_{i2}, i = 1, \ldots, n\right] \\ &= \beta_1 + \frac{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1) E\left[\epsilon_i | x_{i1}, x_{i2}, i = 1, \ldots, n\right]}{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)^2} \\ &= \beta_1 \quad \because \textit{Theorem } 2.2.22 \end{aligned}$$

If $E\left[\hat{\beta}_1 | x_{i1}, x_{i2}, i = 1, \ldots, n\right] = \beta_1$,
what is $E[\hat{\beta}_1]$?

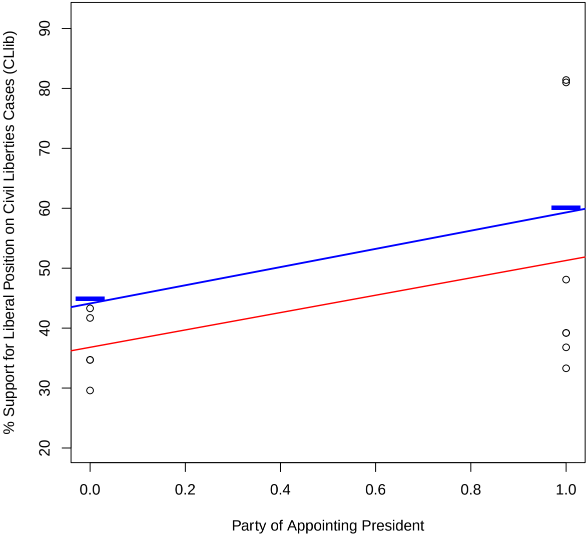
Linear Conditional Mean Function



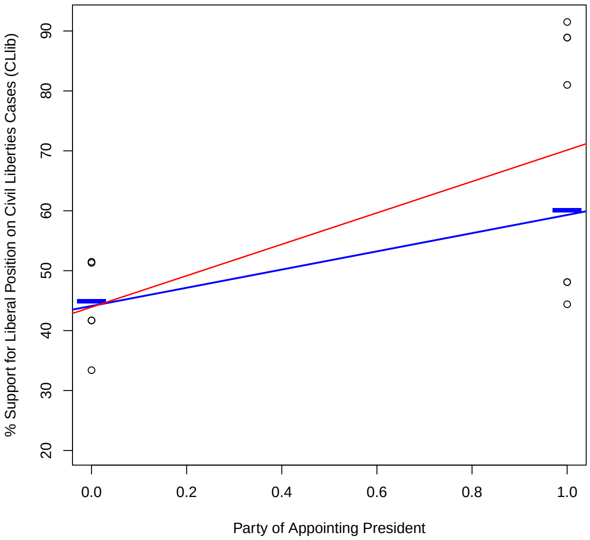
Linear Conditional Expectation Function with Samples



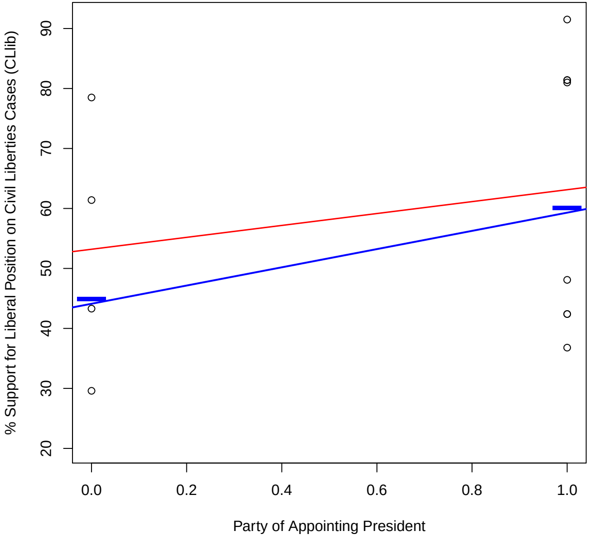
Linear Conditional Expectation Function with Samples



Linear Conditional Expectation Function with Samples



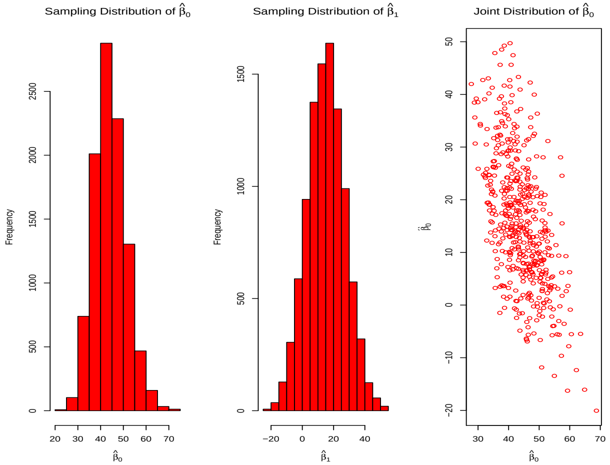
Linear Conditional Expectation Function with Samples



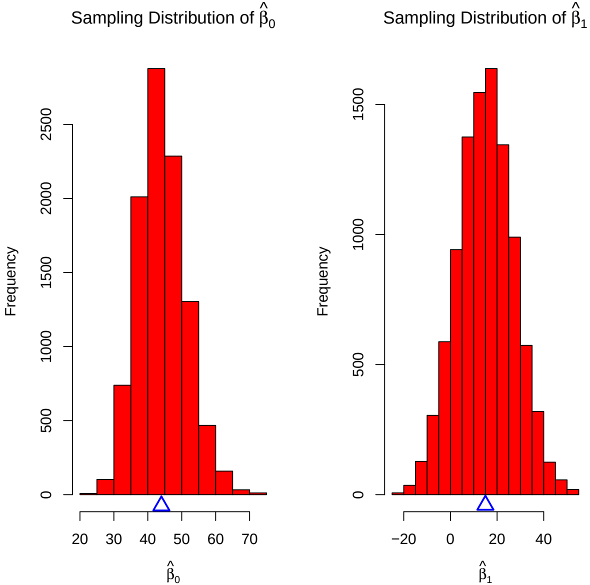
Sampling distribution of the sample slope

i	1	2	...	12	
s	Y_1, X_1	Y_2, X_2	...	Y_{12},X_{12}	$\hat{\beta}_1$
1	35,1	50,0	...	22,1	14.49
2	28,1	19,1	...	65,0	26.21
3	33,1	27,1	...	45,0	9.93
4	$\vdots$	$\vdots$	...	$\vdots$	$\vdots$
5	$\vdots$	$\vdots$	...	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$\mathrm{E}_{\cdot}(\cdot)$	50.9, 0.44	50.9, 0.44	...	50.9, 0.44	15

Sampling Distributions for $\hat{\beta}_0$ and $\hat{\beta}_1$

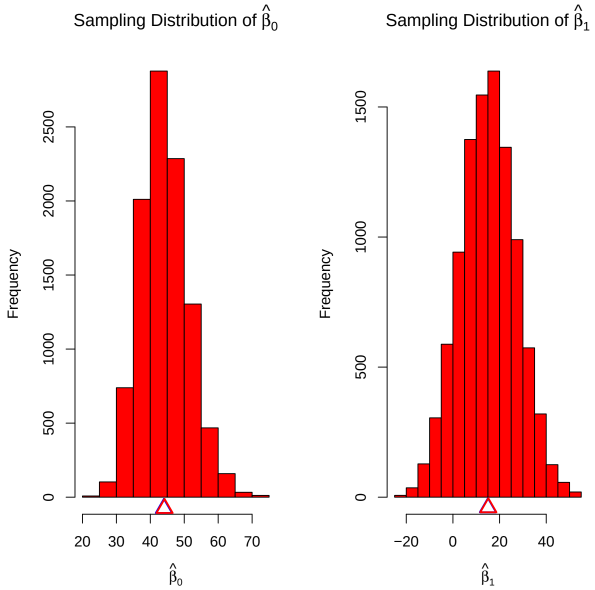


Bias



Sampling Distribution of $\hat{\beta}_0$
Frequency
0
500
1000
1500
2000
2500
20
30
40
50
60
70
$\hat{\beta}_0$
Sampling Distribution of $\hat{\beta}_1$
Frequency
0
500
1000
1500
−20
0
20
40
$\hat{\beta}_1$

Bias



Sampling Distribution of $\hat{\beta}_0$
Frequency
0
500
1000
1500
2000
2500
20
30
40
50
60
70
$\hat{\beta}_0$
Sampling Distribution of $\hat{\beta}_1$
Frequency
0
500
1000
1500
−20
0
20
40
$\hat{\beta}_1$

Assumptions for Unbiasedness

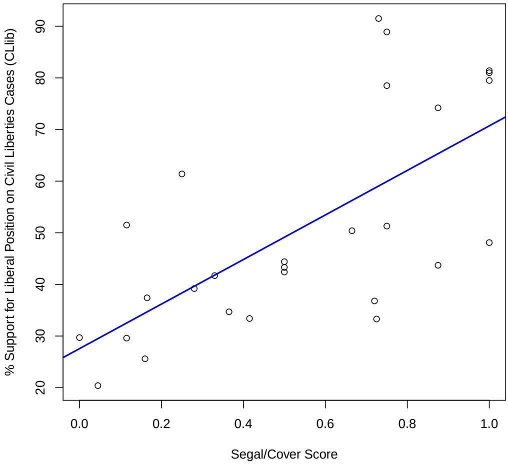
What assumptions did we need for $E[\hat{\beta}_0] = \beta_0$ and $E[\hat{\beta}_1] = \beta_1$?

Assumptions for Unbiasedness

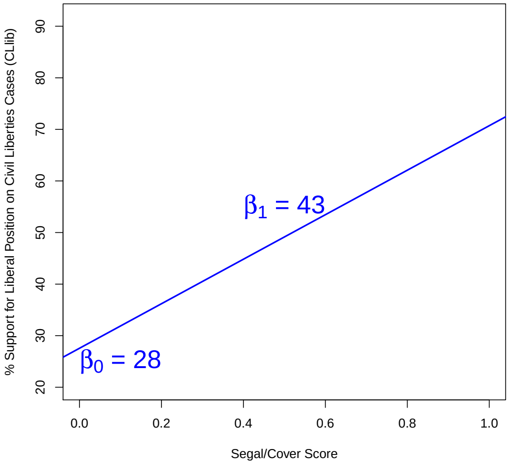
What assumptions did we need for $E[\hat{\beta}_0] = \beta_0$ and $E[\hat{\beta}_1] = \beta_1$?

Just random sampling!

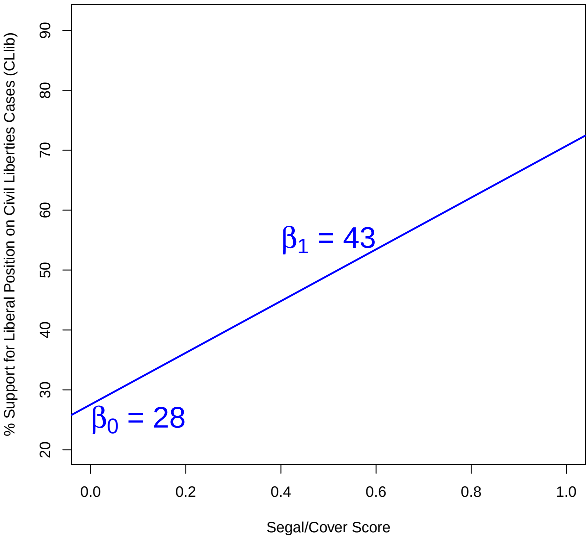
BLP for CLlib Conditional on SCscore



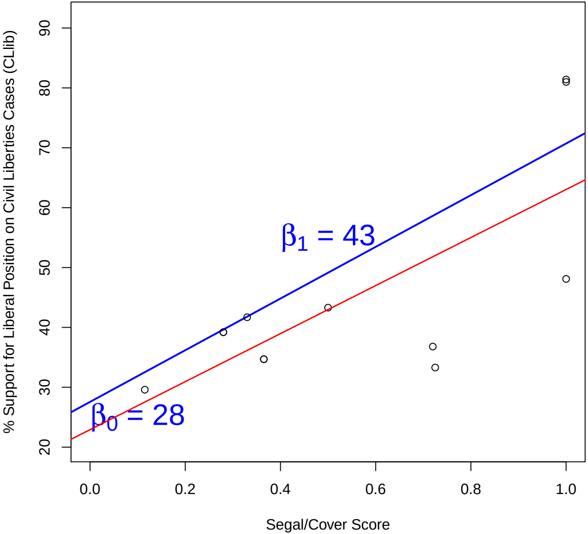
BLP for CLlib Conditional on SCscore



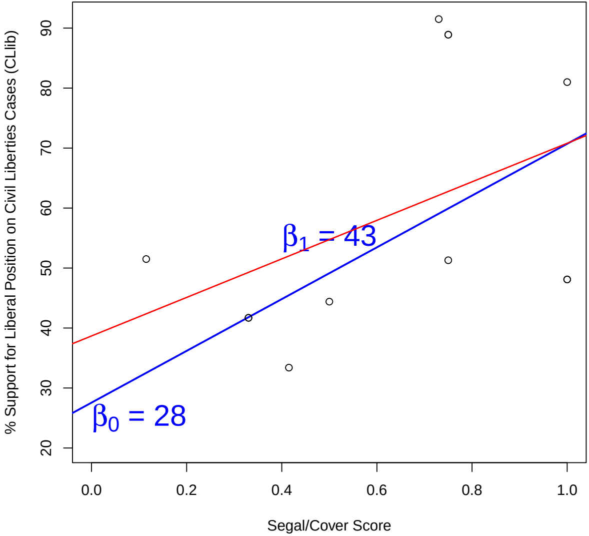
Linear Conditional Expectation Function with Samples



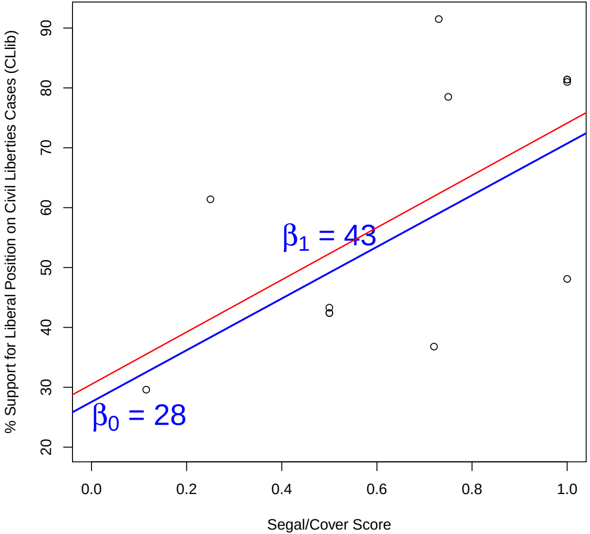
Linear Conditional Expectation Function with Samples



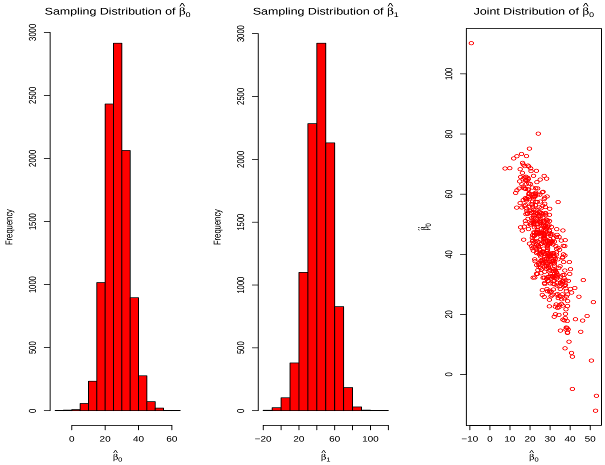
Linear Conditional Expectation Function with Samples



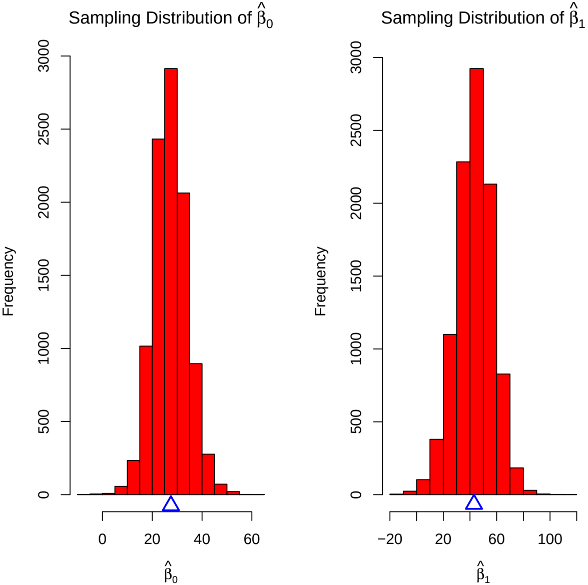
Linear Conditional Expectation Function with Samples



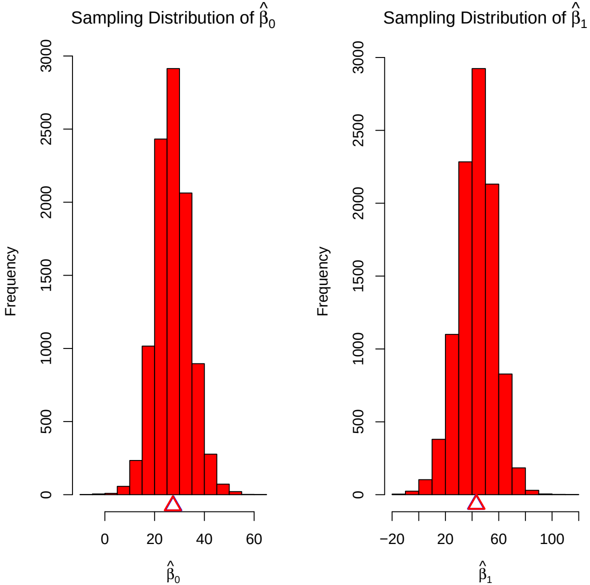
Sampling Distributions for $\hat{\beta}_0$ and $\hat{\beta}_1$



Bias

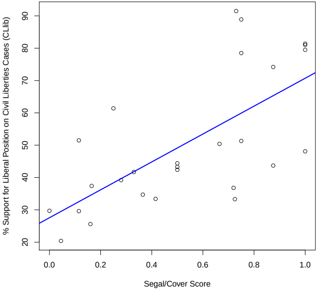

Sampling Distribution of $\hat{\beta}_0$
Frequency
0
500
1000
1500
2000
2500
3000
0
20
40
60
$\hat{\beta}_0$
Sampling Distribution of $\hat{\beta}_1$
Frequency
0
500
1000
1500
2000
2500
3000
−20
20
60
100
$\hat{\beta}_1$

Bias


Sampling Distribution of $\hat{\beta}_0$
Frequency
0
500
1000
1500
2000
2500
3000
0
20
40
60
$\hat{\beta}_0$
Sampling Distribution of $\hat{\beta}_1$
Frequency
0
500
1000
1500
2000
2500
3000
−20
20
60
100
$\hat{\beta}_1$

Assumptions for Unbiasedness

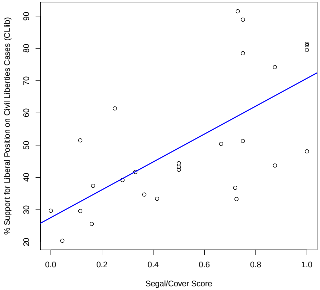
What assumptions did we need for $E[\hat{\beta}_0] = \beta_0$ and $E[\hat{\beta}_1] = \beta_1$?



Assumptions for Unbiasedness

What assumptions did we need for $E[\hat{\beta}_0] = \beta_0$ and $E[\hat{\beta}_1] = \beta_1$?

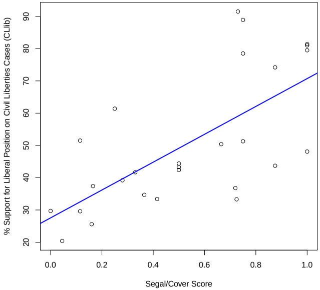
Just random sampling!



Assumptions for Unbiasedness

What assumptions did we need for $E[\hat{\beta}_0] = \beta_0$ and $E[\hat{\beta}_1] = \beta_1$?

Just random sampling!



Unless we really wanted to estimate the CEF and not the BLP.

Consistency of $\hat{\beta}_1$ for β_1

Proving consistency of $\hat{\beta}_1$ for β_1

Recall that,

$$\beta_1 = \frac{C[X, Y]}{V[X]} = \frac{E[YX] - E[Y]E[X]}{E[X^2] - E[X]^2}$$

and the plug in estimator is

$$\hat{\beta}_1 = \frac{\overline{YX} - \overline{Y} \cdot \overline{X}}{\overline{X^2} - \overline{X}^2}.$$

Prove consistency.

Multivariate Regresssion

Define

$$\mathbf{X} = \begin{pmatrix} 1 & X_{11} & X_{21} & \dots & X_{K1} \\ 1 & X_{12} & X_{22} & \dots & X_{K2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & X_{1n} & X_{2n} & \dots & X_{Kn} \end{pmatrix}, \quad \mathbf{Y} = \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix}, \quad \mathbf{e} = \begin{pmatrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{pmatrix},$$

$$\text{and} \quad \widehat{\boldsymbol{\beta}} = \begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \\ \hat{\beta}_2 \\ \vdots \\ \hat{\beta}_K \end{pmatrix}.$$

Then it is equivalent to write the following:

$$
\begin{aligned}
Y_1 &= \hat{\beta}_0 + \hat{\beta}_1 X_{11} + \hat{\beta}_2 X_{21} + ...\hat{\beta}_K X_{K1} + e_1 \\
Y_2 &= \hat{\beta}_0 + \hat{\beta}_1 X_{12} + \hat{\beta}_2 X_{22} + ...\hat{\beta}_K X_{K2} + e_2 \\
&\;...\\
Y_n &= \hat{\beta}_0 + \hat{\beta}_1 X_{1n} + \hat{\beta}_2 X_{2n} + ...\hat{\beta}_K X_{Kn} + e_n
\end{aligned}
$$

Then it is equivalent to write the following:

$$\begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix} = \begin{pmatrix} 1 & X_{11} & X_{21} & \dots & X_{K1} \\ 1 & X_{12} & X_{22} & \dots & X_{K2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & X_{1n} & X_{2n} & \dots & X_{Kn} \end{pmatrix} \begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \\ \hat{\beta}_2 \\ \vdots \\ \hat{\beta}_K \end{pmatrix} + \begin{pmatrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{pmatrix}.$$

Then it is equivalent to write the following:

$$\mathbf{Y} = \mathbf{X}\widehat{\beta} + \mathbf{e}.$$

OLS Solution

$$\widehat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$$

Residuals can be written as the following:

$$\begin{pmatrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{pmatrix} = \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix} - \begin{pmatrix} 1 & X_{11} & X_{21} & \dots & X_{K1} \\ 1 & X_{12} & X_{22} & \dots & X_{K2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & X_{1n} & X_{2n} & \dots & X_{Kn} \end{pmatrix} \begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \\ \hat{\beta}_2 \\ \vdots \\ \hat{\beta}_K \end{pmatrix}.$$

$$\mathbf{e} = \mathbf{Y} - \mathbf{X}\widehat{\boldsymbol{\beta}}.$$

Minimizing the sum of squared residuals:

$$\widehat{\beta} = argmin_{\mathbf{b}}(\mathbf{Y} - \mathbf{X}\mathbf{b})'(\mathbf{Y} - \mathbf{X}\mathbf{b})$$

Setting derivatives equal to zero and substituting $\widehat{\beta}$ for **b.**

$$\begin{aligned} 2\mathbf{X}^T(\mathbf{Y} - \mathbf{X}\widehat{\beta}) &= 0 \\ \mathbf{X}^T\mathbf{e} &= 0 \end{aligned}$$

Let's solve the first equation for $\widehat{\beta}$.

Recall that **Y** can also be written in terms of $\boldsymbol{\beta}$ and ϵ

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \epsilon.$$

Let's draw a picture to remind ourselves of the difference between $\widehat{\boldsymbol{\beta}}$ and $\boldsymbol{\beta}$ and **e** and ϵ.

OLS Solution is Unbiased

$$
\begin{aligned}
\widehat{\boldsymbol{\beta}} &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y} \\
&= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\mathbf{X}\boldsymbol{\beta} + \epsilon) \\
&= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}\boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\epsilon) \\
&= \boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\epsilon
\end{aligned}
$$

$$
\begin{aligned}
E[\widehat{\boldsymbol{\beta}}|\mathbf{X}] &= E[\boldsymbol{\beta}|\mathbf{X}] + E[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\epsilon|\mathbf{X}] \\
&= \boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'E[\epsilon|\mathbf{X}] \\
&= \boldsymbol{\beta} \because \mathit{Theorem} 2.2.22
\end{aligned}
$$

By iterated expectations,

$$
\begin{aligned}
E[\widehat{\beta}] &= E_{\mathbf{X}}\{E[\widehat{\beta}|\mathbf{X}]\} \\
&= E_{\mathbf{X}}\{\boldsymbol{\beta}\} \\
&= \boldsymbol{\beta}
\end{aligned}
$$

Consistency

$$\widehat{\boldsymbol{\beta}} \to_p \boldsymbol{\beta}$$

Handwaving Proof:

- Any of the K slopes in $\boldsymbol{\beta}$ can be rewritten as a simple regression coefficient using the FWL.
- We've previously shown how to prove the consistency of simple regression coefficients using the LLN and CMT.
- There are a few details I'm leaving out here.

Sampling distributions of the simple and multiple sample slopes (in the X direction).

i	1	2	...	12		
s	Y_1, X_1, Z_1	Y_2, X_2, Z_2	...	Y_{12}, X_{12}, Z_{12}	$\hat{\beta}_1^s$	$\hat{\beta}_1^m$
1	3.3,97,100	57.2,36,72.6	...	39.3,30,40.6	-.498	-.499
2	30.4,54,99.3	54.0,24,28.5	...	6.2,99,100	-.456	-.398
3	14.7,87,94.8	3.8,98,100	...	3.8,98,100	-.209	-.190
4	$\vdots$	$\vdots$	...	$\vdots$	$\vdots$	
5	$\vdots$	$\vdots$	...	$\vdots$	$\vdots$	
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	
$\mathrm{E}_\cdot(\cdot)$	17.3, 70.7,89.3	17.3, 70.7,89.3	...	17.3, 70.7,89.3	-.52	-.33

Omitted Variable Bias

- Suppose the model you've estimated is

$$y_i = \beta_0 + \beta_1 x_{i1} + u_i$$

when the model you wanted was

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + u_i$$

- Then your OLS estimator, $\hat{\beta}_1$ is

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)y_i}{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)^2}$$

- Use the rules of conditional expectation to derive the bias.

- Recall that the model we wanted was $y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + u_i$. Replace y_i. Then,

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)y_i}{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)^2}$$

$$=$$

$$=$$

- Recall that the model we wanted was $y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + u_i$. Replace y_i. Then,

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_{i1} - \bar{x_1})y_i}{\sum_{i=1}^{n}(x_{i1} - \bar{x_1})^2}$$

$$= \frac{\sum_{i=1}^{n}(x_{i1} - \bar{x_1})(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + u_i)}{\sum_{i=1}^{n}(x_{i1} - \bar{x_1})^2}$$

$$=$$

- Recall that the model we wanted was $y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + u_i$. Replace y_i. Then,

$$
\begin{aligned}
\hat{\beta}_1 &= \frac{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)y_i}{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)^2} \\
&= \frac{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + u_i)}{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)^2} \\
&= \frac{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)\beta_0}{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)^2} + \frac{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)\beta_1 x_{i1}}{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)^2} + \frac{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)\beta_2 x_{i2}}{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)^2} \\
&\quad + \frac{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)u_i}{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)^2}
\end{aligned}
$$

- Therefore we have

$$\hat{\beta}_1 = \beta_1 + \beta_2 \frac{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)x_{i2}}{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)^2} + \frac{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)u_i}{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)^2}$$

Take expectation conditional on the all the $x_{i1}, x_{i2}, i = \ldots, n$,

$$E\left[\hat{\beta}_1 | x_{i1}, x_{i2}, i = 1, \ldots, n\right]$$

$$=$$

$$=$$

- Therefore we have

$$\hat{\beta}_1 = \beta_1 + \beta_2 \frac{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)x_{i2}}{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)^2} + \frac{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)u_i}{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)^2}$$

Take expectation conditional on the all the $x_{i1}, x_{i2}, i = \ldots, n$,

$$\begin{aligned} & E\left[\hat{\beta}_1 | x_{i1}, x_{i2}, i = 1, \ldots, n\right] \\ &= \beta_1 + \beta_2 \frac{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)x_{i2}}{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)^2} + \frac{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)E\left[u_i | x_{i1}, x_{i2}, i = 1, \ldots, n\right]}{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)^2} \\ &= \end{aligned}$$

- Therefore we have

$$\hat{\beta}_1 = \beta_1 + \beta_2 \frac{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)x_{i2}}{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)^2} + \frac{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)u_i}{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)^2}$$

Take expectation conditional on the all the $x_{i1}, x_{i2}, i = \ldots, n$,

$$E\left[\hat{\beta}_1 | x_{i1}, x_{i2}, i = 1, \ldots, n\right]$$

$$= \beta_1 + \beta_2 \frac{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)x_{i2}}{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)^2} + \frac{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)E\left[u_i | x_{i1}, x_{i2}, i = 1, \ldots, n\right]}{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)^2}$$

$$= \beta_1 + \beta_2 \frac{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)x_{i2}}{\sum_{i=1}^{n}(x_{i1} - \bar{x}_1)^2}$$

- Hence the bias is

$$E\left[\hat{\beta}_1 | x_{i1}, x_{i2}, i = 1, \ldots, n\right] - \beta_1 = \beta_2 \frac{\sum_{i=1}^{n} (x_{i1} - \bar{x}_1) x_{i2}}{\sum_{i=1}^{n} (x_{i1} - \bar{x}_1)^2}$$

Multivariate OVB

- Suppose the model you've estimated is

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \epsilon$$

when the model you wanted was

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\Delta + \epsilon$$

Define

$$\mathbf{X} = \begin{pmatrix} 1 & X_{11} & X_{21} & \dots & X_{K1} \\ 1 & X_{12} & X_{22} & \dots & X_{K2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & X_{1n} & X_{2n} & \dots & X_{Kn} \end{pmatrix}, \quad \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_K \end{pmatrix},$$

$$\mathbf{Z} = \begin{pmatrix} Z_{11} & Z_{21} & \dots & Z_{J1} \\ Z_{12} & Z_{22} & \dots & Z_{J2} \\ \vdots & \vdots & \ddots & \vdots \\ Z_{1n} & Z_{2n} & \dots & Z_{Jn} \end{pmatrix}, \quad \Delta = \begin{pmatrix} \delta_1 \\ \delta_2 \\ \vdots \\ \delta_J \end{pmatrix}.$$

$$\mathbf{Y} = \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix}, \quad \epsilon = \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{pmatrix}.$$

- Then your OLS estimator, $\hat{\boldsymbol{\beta}}$ is

$$
\begin{aligned}
\left[\mathbf{X}^T\mathbf{X}\right]^{-1}\mathbf{X}^T\mathbf{Y} &= \hat{\boldsymbol{\beta}} \\
\left[\mathbf{X}^T\mathbf{X}\right]^{-1}\mathbf{X}^T[\mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\Delta + \epsilon] &= \hat{\boldsymbol{\beta}} \\
\left[\mathbf{X}^T\mathbf{X}\right]^{-1}\mathbf{X}^T\mathbf{X}\boldsymbol{\beta} + \left[\mathbf{X}^T\mathbf{X}\right]^{-1}\mathbf{X}^T\mathbf{Z}\Delta + \left[\mathbf{X}^T\mathbf{X}\right]^{-1}\mathbf{X}^T\epsilon &= \hat{\boldsymbol{\beta}} \\
\boldsymbol{\beta} + \left[\mathbf{X}^T\mathbf{X}\right]^{-1}\mathbf{X}^T\mathbf{Z}\Delta + \left[\mathbf{X}^T\mathbf{X}\right]^{-1}\mathbf{X}^T\epsilon &= \hat{\boldsymbol{\beta}}
\end{aligned}
$$

- Use the rules of conditional expectation to derive the bias.

$$
\begin{aligned}
\mathrm{E}\left[\boldsymbol{\beta} + \left[\mathbf{X}^T\mathbf{X}\right]^{-1}\mathbf{X}^T\mathbf{Z}\Delta + \left[\mathbf{X}^T\mathbf{X}\right]^{-1}\mathbf{X}^T\epsilon\right] &= \mathrm{E}\left[\hat{\beta}|\mathbf{X},\mathbf{Z}\right] \\
\boldsymbol{\beta} + \left[\mathbf{X}^T\mathbf{X}\right]^{-1}\mathbf{X}^T\mathbf{Z}\Delta + \left[\mathbf{X}^T\mathbf{X}\right]^{-1}\mathbf{X}^T\mathrm{E}\left[\epsilon|\mathbf{X},\mathbf{Z}\right] &= \mathrm{E}\left[\hat{\beta}|\mathbf{X},\mathbf{Z}\right] \\
\boldsymbol{\beta} + \left[\mathbf{X}^T\mathbf{X}\right]^{-1}\mathbf{X}^T\mathbf{Z}\Delta &= \mathrm{E}\left[\hat{\beta}|\mathbf{X},\mathbf{Z}\right]
\end{aligned}
$$

Example with $K = 2$ and $J = 1$

$$
\begin{aligned}
Z_1 &= \mathbf{X}\Gamma + \nu \\
&= \begin{pmatrix} 1 & X_{11} & X_{21} \\ 1 & X_{12} & X_{22} \\ \vdots & \ddots & \vdots \\ 1 & X_{1n} & X_{2n} \end{pmatrix} \begin{pmatrix} \hat{\gamma}_0 \\ \hat{\gamma}_1 \\ \hat{\gamma}_2 \end{pmatrix} + \begin{pmatrix} \nu_1 \\ \nu_2 \\ \vdots \\ \nu_n \end{pmatrix}
\end{aligned}
$$

$$
\begin{aligned}
\left[\mathbf{X}^T\mathbf{X}\right]^{-1}\mathbf{X}^T\mathbf{Z}\Delta &= \left[\mathbf{X}^T\mathbf{X}\right]^{-1}\mathbf{X}^T Z_1 \cdot \delta_1 \\
&= \hat{\Gamma} \cdot \delta_1 \\
&= \begin{pmatrix} \hat{\gamma}_0 \\ \hat{\gamma}_1 \\ \hat{\gamma}_2 \end{pmatrix} \cdot \delta_1
\end{aligned}
$$

Bias for β_1 is $\hat{\gamma}_1 \cdot \delta_1$

Example with $K = 2$ and $J = 2$ (Part 1)

$$
\begin{aligned}
Z_1 &= \mathbf{X}\Gamma^1 + \nu^1 \\
&= \begin{pmatrix} 1 & X_{11} & X_{21} \\ 1 & X_{12} & X_{22} \\ \vdots & \ddots & \vdots \\ 1 & X_{1n} & X_{2n} \end{pmatrix} \begin{pmatrix} \hat{\gamma}_0^1 \\ \hat{\gamma}_1^1 \\ \hat{\gamma}_2^1 \end{pmatrix} + \begin{pmatrix} \nu_1^1 \\ \nu_2^1 \\ \vdots \\ \nu_n^1 \end{pmatrix}
\end{aligned}
$$

$$
\begin{aligned}
Z_2 &= \mathbf{X}\Gamma^2 + \nu^2 \\
&= \begin{pmatrix} 1 & X_{11} & X_{21} \\ 1 & X_{12} & X_{22} \\ \vdots & \ddots & \vdots \\ 1 & X_{1n} & X_{2n} \end{pmatrix} \begin{pmatrix} \hat{\gamma}_0^2 \\ \hat{\gamma}_1^2 \\ \hat{\gamma}_2^2 \end{pmatrix} + \begin{pmatrix} \nu_1^2 \\ \nu_2^2 \\ \vdots \\ \nu_n^2 \end{pmatrix}
\end{aligned}
$$

Example with $K = 2$ and $J = 2$ (Part 2)

$$\left[\mathbf{X}^T\mathbf{X}\right]^{-1}\mathbf{X}^T\mathbf{Z}\Delta = \left[\mathbf{X}^T\mathbf{X}\right]^{-1}\mathbf{X}^T Z_1 \cdot \delta_1 + \left[\mathbf{X}^T\mathbf{X}\right]^{-1}\mathbf{X}^T Z_2 \cdot \delta_2$$
$$= \begin{pmatrix} \hat{\gamma}_0^1 \\ \hat{\gamma}_1^1 \\ \hat{\gamma}_2^1 \end{pmatrix} \cdot \delta_1 + \begin{pmatrix} \hat{\gamma}_0^2 \\ \hat{\gamma}_1^2 \\ \hat{\gamma}_2^2 \end{pmatrix} \cdot \delta_2$$

Bias for β_1 is $\hat{\gamma}_1^1 \cdot \delta_1 + \hat{\gamma}_1^2 \cdot \delta_2$

Why do we care about SEs?

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- Inference (e.g., confidence intervals)

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- Efficiency (i.e., lower MSE)

How to make CIs

Two main ideas:

1. Normal Approximation

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1. Normal Approximation
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   1.3 estimate components of variance (and bias if necessary)
   1.4 do the CI algebra
2. Bootstrap

We can decompose the $\hat{\beta}$ into the expected value and the sampling error.

$$
\begin{aligned}
\left[\mathbf{X}^T\mathbf{X}\right]^{-1}\mathbf{X}^T\mathbf{Y} &= \hat{\boldsymbol{\beta}} \\
\left[\mathbf{X}^T\mathbf{X}\right]^{-1}\mathbf{X}^T[\mathbf{X}\boldsymbol{\beta} + \epsilon] &= \hat{\boldsymbol{\beta}} \\
\left[\mathbf{X}^T\mathbf{X}\right]^{-1}\mathbf{X}^T\mathbf{X}\boldsymbol{\beta} + \left[\mathbf{X}^T\mathbf{X}\right]^{-1}\mathbf{X}^T\epsilon &= \hat{\boldsymbol{\beta}} \\
\boldsymbol{\beta} + \left[\mathbf{X}^T\mathbf{X}\right]^{-1}\mathbf{X}^T\epsilon &= \hat{\boldsymbol{\beta}}
\end{aligned}
$$

Thus:

$$
\begin{aligned}
V\left[\hat{\boldsymbol{\beta}}\right] &= V\left[\boldsymbol{\beta} + \left[\mathbf{X}^T\mathbf{X}\right]^{-1}\mathbf{X}^T\epsilon\right] \\
&= V\left[\left[\mathbf{X}^T\mathbf{X}\right]^{-1}\mathbf{X}^T\epsilon\right]
\end{aligned}
$$

Estimation: plug in residuals. This is the “sandwich estimator."

$$\hat{V}\left[\hat{\beta}\right] = \left[\mathbf{X}^T\mathbf{X}\right]^{-1}\mathbf{X}^T\text{diag}\left[(\mathbf{Y}-\mathbf{X}\hat{\beta})^2\right]\mathbf{X}\left[\mathbf{X}^T\mathbf{X}\right]^{-1}.$$

This uses the residuals to *estimate* the relationship between the X variables and the errors.

One note: there are many variants of the variance estimators, that involve premultiplying by, e.g., $\frac{n}{n-K}$ or $\frac{n}{n-K-2}$, etc. These are analogous to the unbiased sample variance vs. plug-in sample variance. It's not so simple with the robust variance estimators (many variants: HC0, HC1, HC2) but they all have the same probability limit.

There is an alternative. “Classical" variance estimation. If we *assume* that $V[\epsilon|X_1, X_2, ..., X_K] = \sigma^2$ for all values of $X_1, X_2, ..., X_K$: homoskedasticity. Then:

$$
\begin{aligned}
nV\left[\hat{\boldsymbol{\beta}}\right] &\xrightarrow{p} n\left[\mathrm{E}\left[\mathbf{X}^T\mathbf{X}\right]\right]^{-1}\mathrm{E}\left[\mathbf{X}^T\mathrm{diag}[\epsilon^2]\mathbf{X}\right]\left[\mathrm{E}\left[\mathbf{X}^T\mathbf{X}\right]\right]^{-1} \\
&= n\sigma^2\left[\mathrm{E}\left[\mathbf{X}^T\mathbf{X}\right]\right]^{-1}\mathrm{E}\left[\mathbf{X}^T\mathbf{I}\mathbf{X}\right]\left[\mathrm{E}\left[\mathbf{X}^T\mathbf{X}\right]\right]^{-1} \\
&= n\sigma^2\left[\mathrm{E}\left[\mathbf{X}^T\mathbf{X}\right]\right]^{-1}\mathrm{E}\left[\mathbf{X}^T\mathbf{X}\right]\left[\mathrm{E}\left[\mathbf{X}^T\mathbf{X}\right]\right]^{-1} \\
&= n\sigma^2\left[\mathrm{E}\left[\mathbf{X}^T\mathbf{X}\right]\right]^{-1}
\end{aligned}
$$

Where $\sigma^2 = V[\epsilon] = \mathrm{E}\left[\epsilon^2\right]$.

Classical standard errors make an additional assumption: the magnitude of the errors is *unrelated* to the values of the Xs.

This assumption is unlikely to hold. Usually – in real life – the errors will be “heteroskedastic," because in real life, data is messy. If homoskedasticity *does* hold, then in large samples, the classical and “robust" variance estimates will agree.

With large n and i.i.d. sampling, always use robust standard errors or bootstrapping to characterize the uncertainty of your regression fit.

If Y and X have been centered and x is a nonstochastic scalar (everything works basically the same if we relax non-stochastic assumption)

$$\begin{aligned} V\left[\hat{\beta}\right] &= V\left[\beta + \left[\frac{\sum_i x_i \epsilon_i}{\sum_i x_i^2}\right]\right] \\ &= \frac{V\left[\sum_i x_i \epsilon_i\right]}{\left[\sum_i x_i^2\right]^2} \end{aligned}$$

How does this further simplify if homoskedasticity?

Bootstrapping

1. Take a sample of size n with replacement from your sample
2. Calculate $\beta^{(s=1)}$ from that bootstrap sample
3. Repeat for $s = 1, ...S$ where S is big
4. Do one of the following:
   - Calculate bootstrapped variance $\frac{1}{S-1}\sum_{s=1}^{S}(\beta^{(s)} - \overline{\beta^{(s)}})^2$
   - Put $\{\beta^{(s)} : s = 1, ..., S\}$ in order and report the $\alpha/2$ and $1 - \alpha/2$ percentiles
   - ...

Takes care of heteroskedasticity automatically.

Standard Errors/Variance (with homoskedasticity)

With homoskedasticity, the simple regression model can be written as:

$$y_i = \beta_0^s + \beta_1^s x_i + \epsilon_i^s$$
$$V[\epsilon_i] = \sigma_s^2$$

and the two covariate multiple regression model can be written as:

$$y_i = \beta_0^m + \beta_1^m x_i + \beta_1^m z_i + \epsilon_i^m$$
$$V[\epsilon_i^m] = \sigma_m^2$$

Comparing variances (with homoskedasticity)

$$V[\hat{\beta}_1^s] = \frac{\sigma_s^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$V[\hat{\beta}_1^m] = \frac{1}{1 - R^2_{x\sim z}} \cdot \frac{\sigma_m^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

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If x is randomized, then what does that tell us about

- β_1^s and β_1^m
- $V[\hat{\beta}_1^s]$ and $V[\hat{\beta}_1^m]$

November 1

Linear Models: what are they good for?

Linear Models

1. Easy to fit
2. Easy to interpret
3. Nice properties (e.g., FWL)
4. Easy SE formulas
5. Easy sensitivity analysis (later)

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1. Easy to fit? (yep, computers)
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5. Easy sensitivity analysis? (later)
6. Robust? (yep, computers and cross-validation)

What if the CEF is nonlinear?

I.e., what if $\mathrm{E}[Y|X] \neq a + bX$?

Can we still use linear regression?

Suppose the data were generated:

$$Y = 10X^2 + U,$$

where X is distributed $\mathit{Uniform}(0, 1)$ and U is distributed $\mathit{Normal}(0, 1)$.

The CEF is nonlinear $\mathrm{E}[Y|X] = 10X^2$. But what would the BLP be?

Draw Figure 4.3.1 on board.

BLP is a good, but not great approximation to the CEF. Can we do better with OLS?

Suppose now we created a second variable: X^2. Then we want to estimate:

$$\underset{\beta_0,\beta_1,\beta_2}{\arg\min}\, \mathrm{E}\,(Y - \beta_0 - \beta_1 X - \beta_2 X^2)^2.$$

The *best linear predictor* with respect to X and X^2 is the CEF here!

$$\text{BLP} = \text{CEF} = \mathrm{E}\left[Y|X, X^2\right] = \mathrm{E}\left[Y|X\right] = \beta_0 + \beta_1 X + \beta_2 X^2.$$

$$\text{BLP} = \text{CEF} = \mathrm{E}[Y|X^2] = \mathrm{E}[Y|X] = 0 + 0X + 10X^2.$$

When the CEF is linear in X, X^2,

$$\mathrm{E}[Y|X] = \beta_0 + \beta_1 X + \beta_2 X^2.$$

When the CEF is not linear in X, X^2,

$$\mathrm{E}[Y|X] \approx \beta_0 + \beta_1 X + \beta_2 X^2.$$

How to interpret slopes?

Before: linear estimate of slope of CEF was β_1.

How can we generalize?

When the CEF is linear in X, X^2,

$$\frac{\partial \mathrm{E}[Y|X]}{\partial X} = \frac{\partial(\beta_0 + \beta_1 X + \beta_2 X^2)}{\partial X}$$

When the CEF is linear in X, X^2,

$$\frac{\partial \mathrm{E}[Y|X]}{\partial X} = \beta_1 + 2\beta_2 X$$

When the CEF is linear in X, X^2,

$$\frac{\partial \mathrm{E}\left[Y|X\right]}{\partial X} = \beta_1 + 2\beta_2 X$$

The slope of the CEF depends on the value of X.

When the CEF is linear in X, X^2,

$$\frac{\partial \mathrm{E}\left[Y|X\right]}{\partial X} = \beta_1 + 2\beta_2 X$$

with

$$g(x) = \beta_0 + \beta_1 x + \beta_2 x^2.$$

what is

$$g(x + 1/2) - g(x - 1/2)$$

Can we develop a general result for approximating the CEF with polynomials? (Yes!)

The Weierstrass approximation theorem:

Any continuous function defined on a closed interval can be uniformly approximated to arbitrary precision using a polynomial function.

Let's visualize this. Let's take a complex CEF:

$$\mathrm{E}\left[Y|X=x\right] = \begin{cases} 0 & : x \in [-1,0] \\ x^2 & : x \in (0,1) \\ 2-x & x \in [1,3] \end{cases}$$

Overfitting

Draw Figures 4.3.4 and 4.3.5 from book

What about multiple X:

- Additive, Interactive, and Saturated Models
- with polynomials
- What if not linear in β?

What happens when the conditional slope of the CEF with respect to X_1 depends on the value of X_2?

The BLP is still the BLP, but we might be missing out on an interesting *interaction* between the variables X_1 and X_2.

So, to formalize a bit, what if the CEF is otherwise linear, but includes an *interaction*?

$$\mathrm{E}[Y|X_1, X_2] = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2$$

Much as before, we could derive the slope of the CEF with respect to a given variable.

$$\frac{\partial \mathrm{E}\left[Y|X_1, X_2\right]}{\partial X_1} = \frac{\partial \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2}{\partial X_1}.$$

Much as before, we could derive the slope of the CEF with respect to a given variable.

$$\frac{\partial \mathrm{E}\left[Y|X_1, X_2\right]}{\partial X_1} = \beta_1 + \beta_3 X_2.$$

$$\frac{\partial \mathrm{E}\left[Y|X_1, X_2\right]}{\partial X_1} = \beta_1 + \beta_3 X_2.$$

The conditional slope of the CEF with respect to X_1 *depends on the value* of X_2.

For full generality, interactions can be combined with polynomials, e.g.:

$$\mathrm{E}\left[Y|X_1, X_2\right] = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2 + \beta_4 X_2^2 + \beta_5 X_1^2 X_2$$

With slopes computed by taking partial derivatives.

Let's find

$$\frac{\partial \mathrm{E}\left[Y|\mathbf{X}\right]}{\partial X_1}|_{\mathbf{X}=x}$$

Average Partial Derivative

The partial derivative can differ for all possible $\mathbf{X} = x$

$$\frac{\partial \mathrm{E}[Y|\mathbf{X}]}{\partial X_1}|_{\mathbf{X}=x}$$

Often, we will be interested in an average of these partials

$$APD_{X_1} = E\left[\frac{\partial \mathrm{E}[Y|\mathbf{X}]}{\partial X_1}|_{\mathbf{X}=x}\right]$$

where the average is taken over the distribution of **X.**

Estimated Average Partial Derivative

In practice, we just average over the partials at the data points

$$\widehat{APD}_{X_1} = \frac{1}{n}\sum_{i=1}^{n}\left[\frac{\partial \mathrm{E}\left[Y|\mathbf{X}\right]}{\partial X_1}|_{\mathbf{X}=x}\right]$$

Estimated APD Examples on board:

Single X examples

- Linear
- Quadratic
- Cubic
- General

X_1,X_2 examples

- Linear Additive
- Linear Interactive
- Polynomial with Interactions

Recall the estimated APD:

$$\widehat{APD}_{X_1} = \frac{1}{n}\sum_{i=1}^{n}\left[\frac{\partial \hat{\mathrm{E}}\left[Y|\mathbf{X}\right]}{\partial X_1}|_{\mathbf{X}=x}\right]$$

What if we had a binary regressor, X_1 (i.e., $\mathrm{Supp}\left(X_1\right) = \{0, 1\}$)?

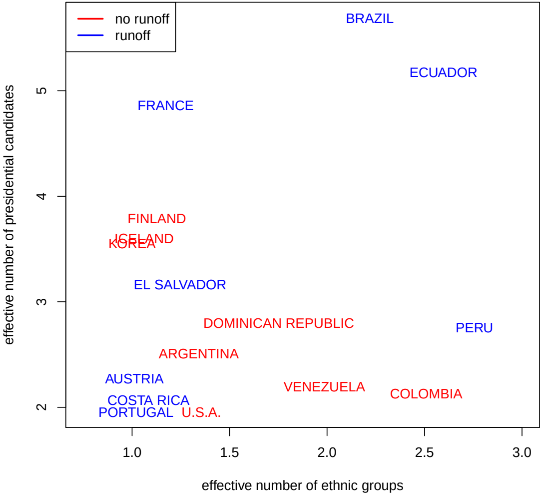
Recall the estimated APD:

$$\widehat{APD}_{X_1} = \frac{1}{n}\sum_{i=1}^{n}\left[\frac{\partial \hat{\mathrm{E}}\left[Y|\mathbf{X}\right]}{\partial X_1}|_{\mathbf{X}=x}\right]$$

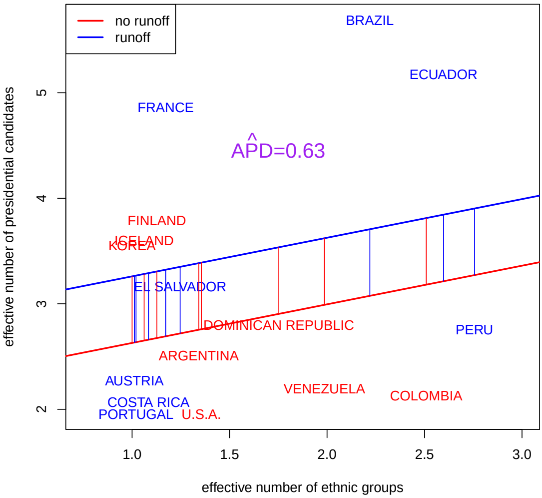
What if we had a binary regressor, X_1 (i.e., $\mathrm{Supp}\left(X_1\right) = \{0, 1\}$)?

$$\widehat{APD}_{X_1} = \frac{1}{n}\sum_{i=1}^{n}\left[\hat{\mathrm{E}}\left[Y|X_1 = 1, \mathbf{X}_{(-1)} = \mathbf{x}_{i,(-1)}\right] - \hat{\mathrm{E}}\left[Y|X_1 = 0, \mathbf{X}_{(-1)} = \mathbf{x}_{i,(-1)}\right]\right]$$

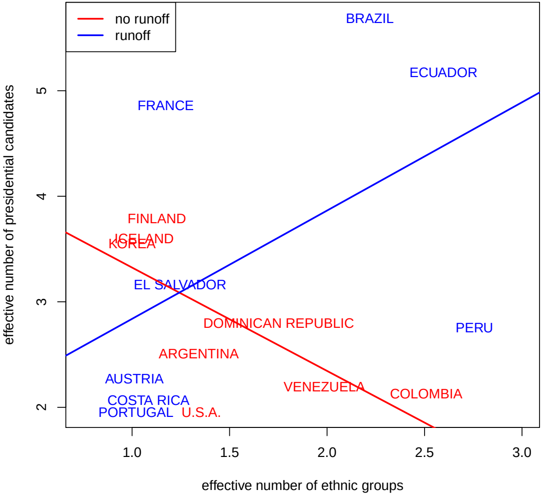
Neto and Cox (1997) Table 3 Data



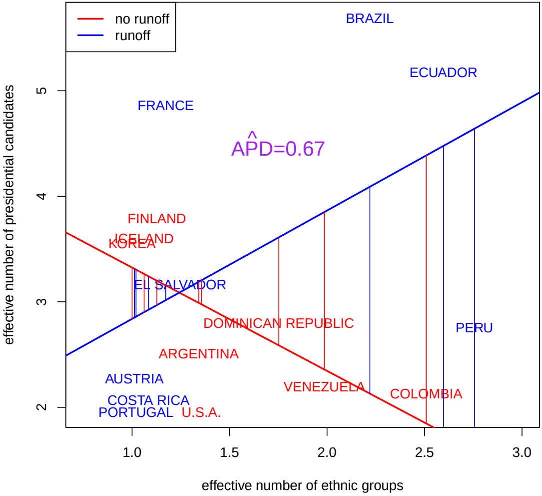
Average Partial Derivative (Runoff) in Additive Model



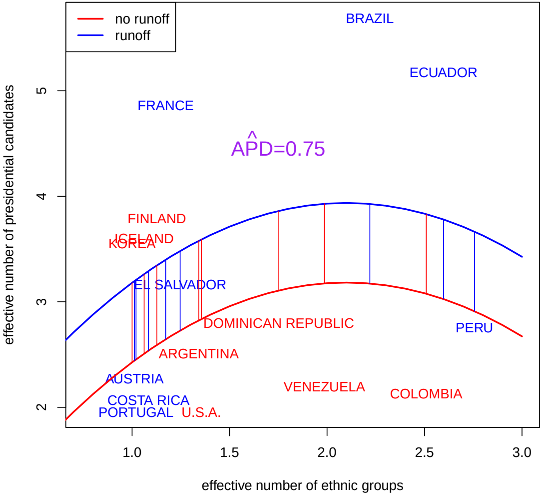
APD in Interactive/Unpooled Linear Model



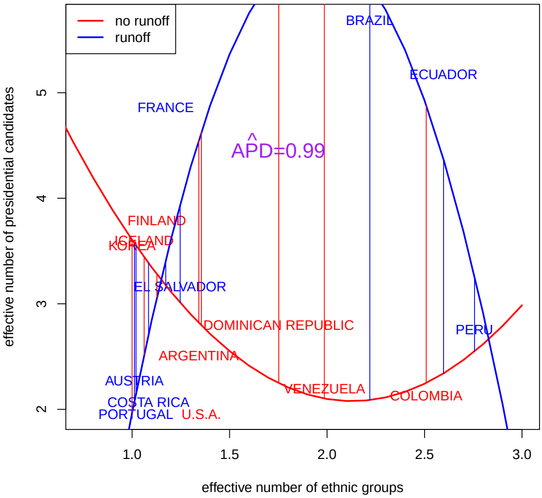
APD in Interactive/Unpooled Linear Model



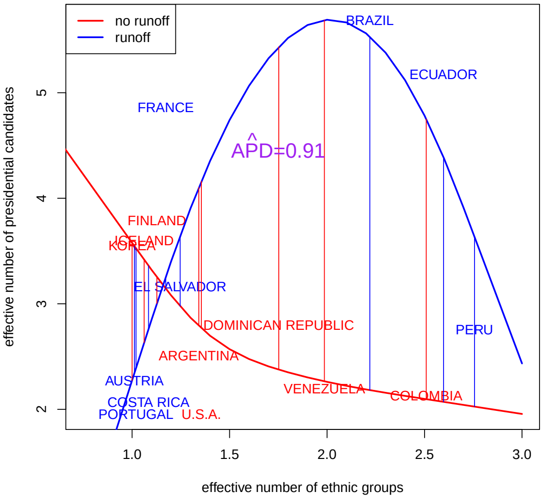
APD in Additive Quadratic Models



APD in Interactive/Unpooled Quadratic



APD in Unpooled Local Regression



Local Regression

- Moving averages (on board)
- go to JWHT ISL book pg. 305

Missing Data (Ch. 6 of AM)

Let's start by considering a simple but common problem: missing data.

What can we learn about the distribution of a variable of interest when some values are missing from, e.g., nonresponse.

E.g., if we ran a survey asking people who they voted for, some people might not respond to the vote choice question. We want to know the population distribution of outcomes to the question, but we don't observe the answers for some people.

Let's start with a minimal assumption. Without much loss of generality, suppose that $\operatorname{Supp}[Y_i] \subset [0, 1]$. (For any bounded outcome variable, we could rescale to $[0, 1]$.)

With this assumption, we can derive *sharp bounds* (bounds that cannot be improved upon without further assumptions) on $\mathrm{E}[Y_i]$.

These bounds are a special case of Manski bounds. See, e.g., Manski (2003).

Manski bounds proceed by imputing the worst-case scenario for all missing data.

For an upper bound, assume that all missing Y_is are a 1. For a lower bound, assume that all missing Y_is are 0.

For example, we might see the following data:

Unit	Y_i^*	R_i
1	1	1
2	-99	0
3	1	1
4	0	1
5	1	1
6	-99	0

Unit	Y_i^*	R_i
1	1	1
2	-99	0
3	1	1
4	0	1
5	1	1
6	-99	0

Unit	Y_i	R_i
1	1	1
2	?	0
3	1	1
4	0	1
5	1	1
6	?	0

For units 2 and 6, we only know that $\mathrm{Supp}\,[Y_i] \subset [0, 1]$.

Using the sample mean as a plug-in estimator for the expected value, our estimate of a lower bound for $\mathrm{E}[Y_i]$:

Unit	Y_i^*	R_i
1	1	1
2	-99	0
3	1	1
4	0	1
5	1	1
6	-99	0

Unit	Y_i	R_i
1	1	1
2	0	0
3	1	1
4	0	1
5	1	1
6	0	0

Estimated lower bound for $\mathrm{E}[Y_i]$ is 1/2.

Using the sample mean as a plug-in estimator for the expected value, our estimate of a upper bound for $\mathrm{E}[Y_i]$:

Unit	Y_i^*	R_i
1	1	1
2	-99	0
3	1	1
4	0	1
5	1	1
6	-99	0

Unit	Y_i	R_i
1	1	1
2	1	0
3	1	1
4	0	1
5	1	1
6	1	0

Estimated upper bound for $\mathrm{E}[Y_i]$ is 5/6.

Go to QoGmissing.R

Ch. 7 of AM will be in different slides.